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Finite Roots of Completely Depolarizing Channels

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CHAPTER 1

Introduction

In the realm of quantum information theory every map that describes the dynamics of a quantum system is referred to as *quantum channel* [Wer00]. The subset of *depolarizing* channels plays an important role as it is used to model noisy evolutions of quantum systems. Therefore, depolarizing channels are of interest in many areas of quantum information such as quantum error correction [CM10], quantum cryptography [JKK10] and quantum computing [BD08]. Representing the dynamics of a quantum system by a depolarizing channel means that there occurs some leakage of knowledge about the initializing state during the evolution of the system. *Completely* depolarizing channels are characterized by the immediate erasure of *all* the information about the input state.

If the input and the output system of a quantum channel are of the same dimension, we can investigate the channel's *divisibility*, i.e. addressing the question of whether a channel can be written as a non trivial concatenation of other channels. The divisibility of quantum channels in general is discussed in [WC08]. If we apply this concept to completely depolarizing channels and restrict the question of divisibility in two ways, namely first the channel should be divided into the concatenation of equal channels and secondly we allow only a finite number of concatenation steps, then we attain the topic of this thesis. We refer to channels as *finite roots* of a completely depolarizing channel, if they split the completely depolarizing channel into a finite number of self-concatenations. Thereby we define the number of concatenation steps as *order* of the root. We investigate how *finely* we can divide the depolarizing process, i.e. the maximal order of a root with respect to the dimension of the underlying system. Furthermore, we introduce approaches to construct maximal finite roots in the context of different classes of channels.

The finite roots of a completely depolarizing channel are applicable in the context of quantum memory channels [KW05] as well as in the context of finitely correlated states [FNW92]. In the case of quantum memory channels we refer to a channel as strictly forgetful if the memory system acts completely depolarizing after some finite steps of concatenation. Therefore a root of a completely depolarizing channel is a strictly forgetful quantum channel without input. In the case of finitely correlated states we use a root of the completely depolarizing channel to define a k-dependent state.

1.1 Overview

After a brief introduction in chapter one, the second chapter deals with the mathematical formalism and notions used throughout this thesis. The concept of quantum channels and their basic representations are introduced, where particular attention is paid to the class of completely depolarizing channels. Moreover, we give a formal definition of a root of a quantum channel. Eventually, we review the theorem of Jordan decomposition, which turns out to be an essential tool to obtain the results of this thesis.

In the third chapter we shift the problem of the determination of finite roots to classical statistical systems. We discuss stochastic matrices due to the fact that they represent classical channels. We present approaches to construct roots of maximal order and put them into the context of Birkhoff's theorem. Finally, we present a way to embed classical roots into quantum systems.

In the fourth chapter we investigate the construction of maximal roots in the most elementary quantum case, i.e. the qubit case. We consider unitary channels as well as Pauli diagonal channels and find that all qubit channels can be written as a composition of both. This allows for the construction of maximal qubit roots. Furthermore, we introduce various approaches to embed such roots into higher dimensional systems.

The fifth chapter is dedicated to a class of Clifford channels, where the dimension of the input and output systems is restricted to be prime. We discuss the basic properties of these Clifford channels and introduce a method to construct higher order Clifford roots. Moreover, we prove that there is a maximum for the order of a Clifford root.

Chapter six covers the construction of maximal roots in arbitrary dimensions. This being the main result of the thesis, the rest of this chapter is concerned with applications of maximal roots in finitely correlated states and quantum memory channels. Chapter seven closes with a short summary of the results and a brief outlook.

CHAPTER 2

Mathematical Framework

It is a well-known fact that quantum mechanics is a statistical theory. We will give a short introduction to the mathematical concepts by which the probabilities in quantum mechanics are described.

2.1 States and Observables

The underlying structure in quantum mechanics is a complex Hilbert space, i.e. a vector space $\mathcal{H}$ over $\mathbb{C}$ equipped with a sesquilinear form $\langle \cdot, \cdot \rangle : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{C}$ where the induced normed vector space is complete. In this thesis we deal exclusively with finite dimensional Hilbert spaces which are isomorphic to $\mathcal{H} = \mathbb{C}^d$ for some $d \in \mathbb{N}$. The set $\mathcal{B}(\mathcal{H})$, which denotes the bounded operators on that Hilbert space, is given by the algebra of complex $d \times d$ matrices (denoted as $M_d(\mathbb{C})$). Before we define the mathematical expression of a *state* of the system we need to introduce the *positivity* of an element of $\mathcal{B}(\mathcal{H})$:

Definition 2.1 (Positivity). An operator $A \in \mathcal{B}(\mathcal{H})$ is called **positive** (we write $A \geq 0$), if it satisfies the following three equivalent conditions:

- $\exists B \in \mathcal{B}(\mathcal{H})$ such that $A = B^*B$
- $A = A^*$ and $\forall \lambda_i \in \text{spec}(A) : \lambda_i \geq 0$
- $\forall \Psi \in \mathcal{H} : \langle \Psi | A \Psi \rangle \geq 0$

The set of possible *states* $\mathcal{S}(\mathcal{H})$ of the system is given by the set of *density operators*, $\rho : \mathcal{H} \rightarrow \mathcal{H}$, $\rho \geq 0$, $\text{tr}(\rho) = 1$. The possible measurements are described by *POVMs*¹, which consist of a set of positive operators F_x corresponding to an element x of the possible set of outcomes $\mathfrak{X}$ and sum up to one: $F_x \geq 0$, $\sum_{x \in \mathfrak{X}} F_x = \mathbf{1}$. The set of operators forms an *observable* X . The *probability* (depicted figure 2.1) of measuring the outcome x for a system in the state ρ can be obtained via $p(x) = \text{tr}(\rho F_x)$. The expectation value of the observable X if measured in state ρ is $\mathbb{E}_\rho(X) = \sum_{x \in \mathfrak{X}} x p(x) = \text{tr}(\rho X)$. An important special case of POVMs are *PVMs*² where the operators F_x satisfy the projection property $F_x^2 = F_x$ and orthogonality $F_x F_y = F_x \delta_{xy}$ in addition to positivity and completeness.

¹ positive operator-valued measures

² projection-valued measure

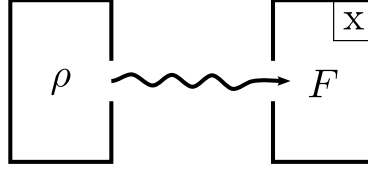


Figure 2.1: The probability to measure the outcome x in a quantum system is completely determined via the state ρ , which represents a preparation of the system, and the corresponding positive operator F_x .

Example 2.1 (Bloch Sphere). We illustrate the set of states in the case $\mathcal{H} = \mathbb{C}^2$. We introduce the Pauli matrices $\{\sigma_0, \sigma_1, \sigma_2, \sigma_3\}$, which form a basis of $\mathcal{B}(\mathbb{C}^2)$:

$$\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (2.1)$$

The hermitian matrices with trace one can be parameterized via:

$$\rho = \frac{1}{2}\sigma_0 + \frac{1}{2}(r_1\sigma_1 + r_2\sigma_2 + r_3\sigma_3) = \frac{1}{2} \begin{pmatrix} 1+r_3 & r_1 - ir_2 \\ r_1 + ir_2 & 1-r_3 \end{pmatrix}, \quad r_1, r_2, r_3 \in \mathbb{R}. \quad (2.2)$$

To assure positivity of ρ its eigenvalues λ_1, λ_2 need to be nonnegative:

$$\det \rho = \lambda_1 \lambda_2 = \frac{1}{4}(1 - r_3^2 - (r_1^2 - (ir_2)^2)) = \frac{1}{4}(1 - r_1^2 - r_2^2 - r_3^2) = \frac{1}{4}(1 - \|\vec{r}\|^2). \quad (2.3)$$

From this we can see that the set of possible states forms a compact and convex set, a 3-dimensional unit sphere (see figure 2.2), in which the coordinates can be interpreted as weights of Pauli matrices.

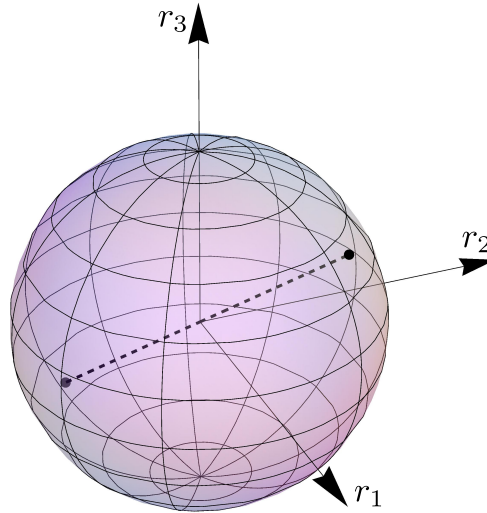


Figure 2.2: The set of possible states can be visualized by the Bloch sphere. Furthermore a PVM is represented by an antipodal pair of points.

The extremal points of this convex set of states are represented by the surface of the sphere and correspond to the *pure* states. The points in the interior of the sphere characterize *mixed* states.

2.2 Channels

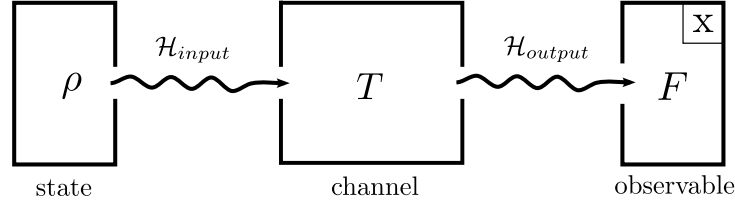


Figure 2.3: Complete quantum system involving a channel. The input and output systems do not necessarily coincide.

Of course, preparations and measurements are not the only possible operations on a quantum system. One could put it in contact with another system or measure a subsystem of a composite system. This leads to the concept of *quantum channels*, which is the most general way to describe the evolution of a quantum system (depicted in figure 2.3). First we give a clear definition of quantum channels:

Definition 2.2 (Quantum Channel). A **quantum channel** is a *linear, completely positive* and in the Heisenberg picture *unital* or in the Schrödinger picture *trace-preserving* map.

The motivation for these properties and their meanings are explained in the following sections.

Heisenberg and Schrödinger picture

The Heisenberg and Schrödinger picture are mathematically equivalent descriptions of quantum mechanics.

In the Heisenberg picture, the channels map the observables of the output system to the observables of the input system:

$$T : \mathcal{B}(\mathcal{H}_{out}) \mapsto \mathcal{B}(\mathcal{H}_{in}). \quad (2.4)$$

In the case of quantum channels, the choice of the Schrödinger picture means that the input algebra of the channel is the state space of the input system,

$$T^* : \mathcal{S}(\mathcal{H}_{in}) \mapsto \mathcal{S}(\mathcal{H}_{out}), \quad (2.5)$$

and thus the channel "follows the system". The star denotes the *dual* operation. Due to the fact that both pictures are equivalent and therefore the probability of outcomes need to be the same, they are connected via the duality relation:

$$\text{tr}(T^*(\rho)F) = \text{tr}(\rho T(F)). \quad (2.6)$$

Throughout this thesis both pictures will be used depending on the convenience of the perspective, but we always denote T as the channel in the Heisenberg and T^* the channel in the Schrödinger picture. We will illustrate this with an example of dual channels.

Example 2.2 (Time Evolution). Let $T^*(\rho) = U_t \rho U_t^*$ be the time evolution of the state ρ for a time-independent hamiltonian. The corresponding channel in the Heisenberg picture can be obtained via:

$$\begin{aligned} \text{tr}(T^*(\rho)F) &= \text{tr}(U_t \rho U_t^* F) = \text{tr}(\rho U_t^* F U_t) = \text{tr}(\rho T(F)) \\ &\Rightarrow T(F) = U_t^* F U_t. \end{aligned}$$

2.2.1 Axiomatic Approach and Complete Positivity

In the axiomatic approach we deduce the properties of a channel as the minimal requirements for a statistical interpretation. In this way we consider figure 2.3 as the setting for a statistical experiment, where we can gather some classical statistical data. In such a case the action of the channel should be determined by all probabilities that can be gathered. If we prepare the input system in a state which is represented by a density operator ρ , the probabilities for any observable X are determined via $\text{tr}[T(\rho)X]$. Therefore we can gain the complete statistical data of the experiment by sweeping through all possible input states. Thus the transformation on the input state is fully characterized by $T^* : \mathcal{S}(\mathcal{H}_{in}) \mapsto \mathcal{S}(\mathcal{H}_{out})$ which maps states to states. This implies that the map needs at least to be trace-preserving,

$$\text{tr}_{\mathcal{H}_{in}}(\rho) = \text{tr}_{\mathcal{H}_{out}}(T^*(\rho)), \quad (2.7)$$

positive,¹

$$\rho \geq 0 \Rightarrow T(\rho) \geq 0, \quad (2.8)$$

and linear, since a mixture of two different preparation-states should result into a mixture of corresponding probabilities.

In the Heisenberg picture, the property of being trace-preserving changes into unitality, i.e. $T(\mathbb{1}) = \mathbb{1}$. However, it turns out that positivity (which is equal in both pictures) is surprisingly not strong enough to ensure that the channels fit into a statistical interpretation.

Completely Positive Maps

There should be no difference in the mathematical properties of a channel, if one composes the channel with a larger system where there is no interaction between each other. This situation is depicted in figure 2.4. Therefore we define the tensor stabilized version of positivity as *complete positivity*:

¹ This is not to be confused with definition 2.1 of positive operators. Here we refer to the positivity of mappings between operators.

Definition 2.3 (Complete Positivity). Let $T : \mathcal{B}(\mathcal{H}_2) \rightarrow \mathcal{B}(\mathcal{H}_1)$ be a linear map. T is called **completely positive** if and only if

$$T \otimes id_n : \mathcal{B}(\mathcal{H}_2) \otimes M_n(\mathbb{C}) \rightarrow \mathcal{B}(\mathcal{H}_1) \otimes M_n(\mathbb{C}) \quad (2.9)$$

takes positive elements to positive elements for all n . In this case id_n denotes the identity (or noiseless) channel defined via

$$id_n : M_n(\mathbb{C}) \rightarrow M_n(\mathbb{C}), \quad id_n(X) = X \quad \forall X \in M_n(\mathbb{C}). \quad (2.10)$$

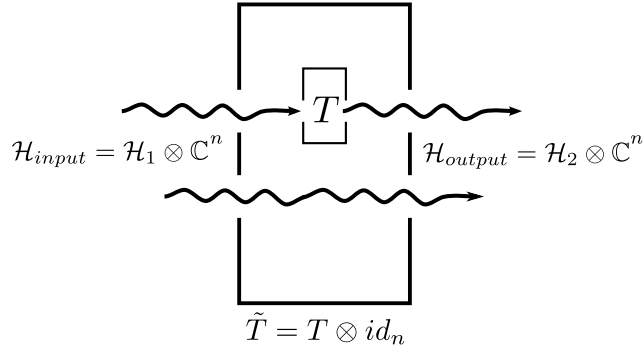


Figure 2.4: The channel T acts only on a part of a composite system.

To introduce some criteria for complete positivity it is useful to consider the positivity of elements of $\mathcal{B}(\mathcal{H}_1) \otimes M_n(\mathbb{C})$. Every operator $A \in \mathcal{B}(\mathcal{H}_1) \otimes M_n(\mathbb{C})$ can be written in the following way:

$$A = \sum_{ij,kl} b_{i,j} |i\rangle \langle j| \otimes c_{k,l} |k\rangle \langle l| = \sum_{kl} \underbrace{\sum_{ij} c_{k,l} b_{i,j} |i\rangle \langle j|}_{A_{k,l}} \otimes |k\rangle \langle l| = \sum_{kl} A_{k,l} \otimes |k\rangle \langle l|. \quad (2.11)$$

This is a block matrix with elements $A_{kl} \in \mathcal{B}(\mathcal{H}_1)$, i.e.

$$A = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} & \dots & \mathbf{A}_{1n} \\ \mathbf{A}_{21} & \mathbf{A}_{22} & \dots & \mathbf{A}_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ \mathbf{A}_{n1} & \mathbf{A}_{n2} & \dots & \mathbf{A}_{nn} \end{pmatrix}. \quad (2.12)$$

The following lemma characterizes the positivity of these block matrices:

Lemma 2.1 ([Pau03]). Let $A = \sum_{kl} A_{kl} \otimes |k\rangle \langle l| \in \mathcal{B}(\mathcal{H}_1) \otimes M_n(\mathbb{C})$. Then A is positive if and only if each A_{kl} can be written in the form $A_{kl} = \sum_{\alpha} (c_k^* c_l)^{\alpha}$, where $c_1, \dots, c_n \in \mathcal{B}(\mathcal{H}_1)$ and $\alpha \in \{1, \dots, n\}$.

This lemma leads us to the following characterization of complete positivity:

Lemma 2.2 ([HW07]). A channel $T : \mathcal{B}(\mathcal{H}_2) \mapsto \mathcal{B}(\mathcal{H}_1)$ is completely positive if and only if $\sum_{ij} a_i^* T[b_i^* b_j] a_j \geq 0$ for all $a \in \mathcal{B}(\mathcal{H}_1)$ and $b \in \mathcal{B}(\mathcal{H}_2)$.

As preparation for the proof we consider the following equivalences:

$$\begin{aligned}
\text{T is completely positive} &\Leftrightarrow \forall b \in \mathcal{B}(\mathcal{H}_2) : \sum_{ij} T \otimes id_n(b_i^* b_j \otimes |i\rangle \langle j|) \geq 0 \\
&\Leftrightarrow \forall b \in \mathcal{B}(\mathcal{H}_2) : \sum_{ij} T[b_i^* b_j] \otimes |i\rangle \langle j| \geq 0 \\
&\Leftrightarrow \exists c \in \mathcal{B}(\mathcal{H}_1) \forall b \in \mathcal{B}(\mathcal{H}_2) : T[b_i^* b_j] = \sum_{\alpha} (c_i^* c_j)^{\alpha},
\end{aligned} \tag{2.13}$$

where we used lemma 2.1 for the second and the last equivalence.

Proof. Now let T be completely positive, then it follows:

$$\sum_{i,j} a_i^* T[b_i^* b_j] a_j = \sum_{i,j,\alpha} a_i^* (c_i^* c_j)^{\alpha} a_j = \sum_{\alpha} X_{\alpha}^* X_{\alpha} \geq 0 \quad \text{with} \quad X_{\alpha} = \sum_i c_i^{\alpha} a_i. \tag{2.14}$$

Conversely we take some invertible $a_i, a_j \in \mathcal{B}(\mathcal{H}_1)$ and see

$$\begin{aligned}
a_i^* T[b_i^* b_j] a_j \geq 0 &\Leftrightarrow \exists d \in \mathcal{B}(\mathcal{H}_1) : a_i^* T[b_i^* b_j] a_j = d^* d \\
&\Leftrightarrow T[b_i^* b_j] = (a_i^{-1})^* d^* d a_j^{-1} = c_i^* c_j \quad \text{with} \quad c_i = d a_i^{-1}.
\end{aligned}$$

□

Subsequently, we give an example of a completely positive map and a map, which is positive but not completely positive:

Example 2.3 (Kraus Map). For any operator $K : \mathcal{H}_1 \rightarrow \mathcal{H}_2$ the map

$$T : \mathcal{B}(\mathcal{H}_2) \mapsto \mathcal{B}(\mathcal{H}_1), \quad T(X) = K^* X K \tag{2.15}$$

is completely positive. It maps positive elements to positive elements, i.e.

$$X \geq 0 \Leftrightarrow \langle \Psi | X \Psi \rangle \geq 0 \Rightarrow \langle \Psi | K^* X K \Psi \rangle = \langle K \Psi | X | K \Psi \rangle \geq 0, \tag{2.16}$$

for all $\Psi \in \mathcal{H}_1$. The same argument holds for complete positivity, if we consider $T \otimes id_n$ equipped with $K \otimes id_n$ instead of K .

Example 2.4 (Pancake Map). Since the state space in the qubit case ($\mathcal{H} = \mathbb{C}^2$) can be visualized by the Bloch sphere, a qubit channel needs to be an affine mapping of the Bloch sphere. Let us consider the map

$$T^* : \mathcal{S}(\mathbb{C}^2) \rightarrow \mathcal{S}(\mathbb{C}^2), \quad T^*(X) = \text{tr}(\mathbb{1}X)\mathbb{1} + \text{tr}(\sigma_2 X)\sigma_2 + \text{tr}(\sigma_3 X)\sigma_3. \tag{2.17}$$

This map projects the Bloch sphere to a Bloch circle with $T^*(\mathbb{1}) = \mathbb{1}$, $T^*(\sigma_2) = \sigma_2$ and $T^*(\sigma_3) = \sigma_3$ and is thus positive, but to satisfy complete positivity an additional deformation of that circle into an ellipse is necessary as we will see later in section 4.2.

2.3 Representations of Channels

In this section we introduce the most important methods for representing a quantum channel. We begin with the channel-state duality that is a useful tool for characterizing the set of possible channels. From this we derive the Kraus decomposition and Stinespring dilation of a channel. We close this section with channels being represented as matrices due to the fact that they are linear maps.

2.3.1 Choi-Jamiolkowski Correspondence

First we define the Choi matrix of a map T .

Definition 2.4 (Choi Matrix). For the linear map $T : \mathcal{B}(\mathcal{H}_2) \rightarrow \mathcal{B}(\mathcal{H}_1)$ the **Choi matrix** $\sigma_T \in \mathcal{B}(\mathcal{H}_1 \otimes \mathcal{H}_2)$ is given by $\sigma_T = T \otimes id_n(|\Omega\rangle\langle\Omega|) = \frac{1}{d} \sum_{i,j} T(|i\rangle\langle j|) \otimes |i\rangle\langle j|$ with $|\Omega\rangle = \frac{1}{\sqrt{d}} \sum_i |ii\rangle$ as a maximally entangled state of dimension $d = \dim \mathcal{H}_2$.

The bijective map $T \rightarrow \sigma_T$ and its inverse give an important connection between channels and states, which we can derive from the following theorem.

Theorem 2.1 (Choi's Theorem [Cho75]). *The map $T : \mathcal{B}(\mathcal{H}_2) \rightarrow \mathcal{B}(\mathcal{H}_1)$ is completely positive, if and only if the corresponding Choi matrix is positive.*

We sketch the proof of this theorem [HW07]:

For any linear map $T : \mathcal{B}(\mathcal{H}_2) \mapsto \mathcal{B}(\mathcal{H}_1)$ we evaluate,

$$\langle i\alpha | \sigma_T | j\beta \rangle = \frac{1}{d} \langle i\alpha | \sum_{i,j} T(|i\rangle\langle j|) \otimes |i\rangle\langle j| | j\beta \rangle = \frac{1}{d} \langle i | T(|\alpha\rangle\langle\beta|) | j \rangle. \quad (2.18)$$

Therefore a linear map $T : \mathcal{B}(\mathcal{H}_2) \mapsto \mathcal{B}(\mathcal{H}_1)$ is uniquely determined by the coefficients of a Choi matrix $\sigma_T \in \mathcal{B}(\mathcal{H}_1 \otimes \mathcal{H}_2)$ and vice versa. If the map T is completely positive, then by definition the image of a maximally entangled state is also a state and thus positive. However, if we first consider the case that σ_T is of the form $\sigma_T = |A\rangle\langle A|$ for $|A\rangle = \sum_{i,\alpha} C_{i,\alpha} |i\alpha\rangle \in \mathcal{H}_1 \otimes \mathcal{H}_2$,

$$\langle i\alpha | \sigma_T | j\beta \rangle = \frac{1}{d} \overline{C_{i\alpha}} C_{j\beta} = \langle \alpha | K(|i\rangle\langle j|) K^* | \beta \rangle, \quad (2.19)$$

with $\langle \beta | K | j \rangle = C_{j\beta} \sqrt{d}$. It follows that T is of the form $T(X) = K^* X K$, which induces complete positivity. Due to spectral decomposition, σ_T can be written as a sum of such terms $\sigma_T = \sum_i \lambda_i |A_i\rangle\langle A_i|$ with $\lambda_i \geq 0$ and because of the linearity of this relation it follows that T is also a sum of positive weighted completely positive terms, which induces complete positivity for T .

We now give an overview of additional relations between a linear map T and its corresponding Choi matrix σ_T :

$$\text{hermiticity : } \sigma_T = \sigma_T^* \Leftrightarrow \forall X \in \mathcal{B}(\mathcal{H}_{in}) : T(X^*) = T(X)^* \quad (2.20)$$

$$\text{unitality : } \text{tr}_{\mathcal{H}_2}(\sigma_T) = \mathbb{1}/d \Leftrightarrow T(\mathbb{1}) = \mathbb{1} \quad (2.21)$$

$$\text{trace-preserving : } \text{tr}_{\mathcal{H}_1}(\sigma_T) = \mathbb{1}/d \Leftrightarrow \forall X \in \mathcal{B}(\mathcal{H}_{in}) : \text{tr}(X) = \text{tr}(T(X)) \quad (2.22)$$

Hence a channel corresponds to a Choi matrix, which is positive and has the maximally mixed state as reduced density operator. The decomposition of T into a sum of completely positive maps leads us to the next section.

2.3.2 Kraus Decomposition and Stinespring Dilation

The Choi matrix is one way to characterize a channel, another useful representation is its Kraus decomposition:

Theorem 2.2 (Kraus Decomposition [Kra83]). *A linear map $T : \mathcal{B}(\mathcal{H}_2) \rightarrow \mathcal{B}(\mathcal{H}_1)$ is completely positive, if and only if it admits a decomposition into Kraus operators $\{K_i\}$ of the following form:*

$$T(X) = \sum_{i=1}^r K_i^* X K_i. \quad (2.23)$$

The additional relationships between T and $\{K_i\}$ are the following:

- T is unital, iff $\sum_{i=1}^r K_i^* K_i = \mathbb{1}$
- T is trace-preserving, iff $\sum_{i=1}^r K_i K_i^* = \mathbb{1}$

The same statement holds for the linear map T^* in the Schrödinger picture, if one interchanges K_i and K_i^* . Notice that the Kraus decomposition is not unique. Two sets of Kraus operators $\{K_i\}$ and $\{\tilde{K}_j\}$ represent the same channel if and only if there exists a unitary matrix U such that $K_i = \sum_j U_{ij} \tilde{K}_j$. Thus the equivalent Kraus decompositions do not necessarily have the same number of Kraus operators. We define the minimal number of Kraus operators as the *Kraus rank* ($\text{rank}(\rho_T)$) of the map T . The Kraus rank is also equivalent to the rank of the Choi matrix, but different from the rank of the map T . For example the ideal channel $T(X) = X$ has full rank as a map, however, the corresponding Choi matrix has rank one.

Example 2.5 (Projective Measurement). We defined a measurement as a set of positive operators F_x which sum up to one. Therefore we can find a close relationship between channels and measurements, if we consider the square roots¹ of the operators as Kraus operators of a channel. The Kraus decomposition of the projective qubit measurement corresponding to the two antipodal points $\{\{0,0,1\}, \{0,0,-1\}\}$ of the Bloch sphere is:

$$T(X) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} X \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} X \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \quad (2.24)$$

Another important representation of channels is given by the *Stinespring representation*:

Theorem 2.3 (Stinespring Representation [Sti55]). *Let $T : \mathcal{B}(\mathcal{H}_2) \rightarrow \mathcal{B}(\mathcal{H}_1)$, $\dim \mathcal{H}_2 = d$ and $\dim \mathcal{H}_1 = d'$. Then there is a number k and an operator $V : \mathbb{C}^{d'} \rightarrow \mathbb{C}^d \otimes \mathbb{C}^k$ such that*

$$T(X) = V^*(X \otimes \mathbb{1}_k)V \quad \forall X \in \mathcal{B}(\mathcal{H}_2), \quad (2.25)$$

¹ Functional analysis shows that for a positive operator A a positive root $B = \sqrt{A}$ is uniquely determined.

where $\mathbb{C}^k$ is called the dilation space and V is an isometry, i.e. $V^*V = \mathbb{1}_{d'}$.

This representation can be obtained from the Kraus decomposition $T(X) = \sum_{i=1}^r K_i^* X K_i$ by constructing $V := \sum_i K_i \otimes |i\rangle$ with $\{|i\rangle\}$ any orthonormal basis in $\mathbb{C}^k$. Since the Kraus decomposition is not unique, the Stinespring dilation has the same unitary degree of freedom. Dilations with $k = \text{rank}(\sigma_T)$ are called *minimal*. Furthermore, the unitality condition in the Kraus decomposition is reflected in V being an isometry. An interesting interpretation arises if we consider the theorem for the channel T^* in the Schrödinger picture via the duality relation

$$\text{tr}(\rho T(X)) = \text{tr}(\rho V^*(X \otimes \mathbb{1}_k)V) = \text{tr}(\text{tr}_{\mathbb{C}^k}(V\rho V^*)X) = \text{tr}(T^*(\rho)X). \quad (2.26)$$

This is the so-called *ancilla representation* of the channel $T^*(\rho) = \text{tr}_{\mathbb{C}^k}(V\rho V^*)$. Since $V : \mathbb{C}^{d'} \rightarrow \mathbb{C}^d \otimes \mathbb{C}^k$ is an isometry it can be expanded to a unitary operator $U : \mathbb{C}^{d'} \otimes \mathbb{C}^{k'} \rightarrow \mathbb{C}^d \otimes \mathbb{C}^k$ where the input state of the system is fixed $V^*\rho V = U^*(\rho \otimes |\phi_0\rangle\langle\phi_0|)U$. In figure 2.5 we illustrate the physical interpretation of this representation. In this picture the dilation space reflects a coupling to the environment.

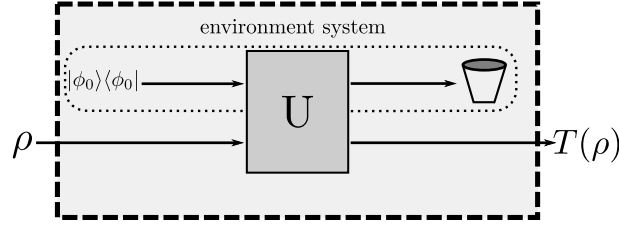


Figure 2.5: Stinespring's representation theorem in the Schrödinger picture. Every quantum channel can be described as a unitary evolution on a bigger system, where the input is composed with an environmental system. The partial trace over the environment after the unitary evolution is represented by the waste bin.

After introducing these representations of channels according to mathematical properties of completely positive maps, we now consider the representation of a channel as matrix due to their linearity.

2.3.3 Channels as Matrices

As mentioned in section 2.1 in the finite dimensional case the set of bounded operators $\mathcal{B}(\mathcal{H})$ can be identified with the complex $d \times d$ matrices $M_d(\mathbb{C})$. This set is closed under linear combinations and scalar multiplication and is therefore a complex vector space of dimension d^2 which can be bijectively mapped to $\mathbb{C}^{d^2}$. If we equip $M_d(\mathbb{C})$ with the scalar product¹

$$\langle A, B \rangle := \text{tr}(A^* B) \quad \forall A, B \in M_d(\mathbb{C}), \quad (2.27)$$

it becomes a Hilbert space. Considering the space of linear maps $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ we find that it is isomorphic to $M_{d'^2, d^2}(\mathbb{C})$. We obtain a concrete representation $\tilde{T}$ in

¹ This is commonly known as the Hilbert-Schmidt scalar product.

orthonormal bases $\{A_1, \dots, A_{d^2}\}$ and $\{B_1, \dots, B_{d^2}\}$ of the input and output space via

$$\tilde{T}_{i,j} = \text{tr}(B_i^* T(A_j)). \quad (2.28)$$

If the map T represents a channel the concatenation is related to the multiplication by the corresponding matrix. The adjoint matrix $\tilde{T}^*$ represents the channel T^* in the dual picture.

Given the equality of input and output algebra, we can use the familiar expressions from linear algebra like eigenvalues, eigenbasis, Jordan decomposition, etc. to characterize a channel. The simplest orthonormal basis with respect to the Hilbert-Schmidt scalar product is the basis consisting of the *matrix units* $\{E_{i,j} := |i\rangle\langle j| \mid i, j \in \{1, \dots, d\}\}$. Using the isomorphism $M_d(\mathbb{C}) \simeq \mathbb{C}^{d^2}$ via $|i\rangle\langle j| \leftrightarrow |i,j\rangle$ it follows that for a channel with Kraus decomposition $T(X) = \sum_i K_i^* X K_i$ the representation matrix is given by $\tilde{T} = \sum_i \overline{K_i} \otimes K_i$.

2.4 Case Study: Completely Depolarizing Channels (CDCs)

The class of depolarizing channels models quantum systems where noise affects the evolution. One common example of depolarizing is given by the situation of sending photons from Alice to Bob via a fiber. If Alice sends some purely prepared photons through a fiber there is a certain probability for the photons to be scattered, such that Bob will measure a mixed state. The loss of information about the input state caused by scattering can be modeled via a convex combination of the ideal channel and a noisy channel:

$$T(X) = (1 - p)X + p \frac{\mathbb{1}_d}{d} \text{tr} X, \quad p \in [0, 1]. \quad (2.29)$$

The parameter p can be understood as a measure for the source of noise in the process represented by the channel T . If we choose $p = 1$ we get a *completely depolarizing channel*¹

$$P(X) = \frac{\mathbb{1}_d}{d} \text{tr} X. \quad (2.30)$$

This channel has rank one and maps an arbitrary input operator to the identity operator multiplied by a constant. The term "completely depolarizing" arises from the representation of the channel in the Schrödinger picture:

$$\text{tr}(\rho P(X)) = \text{tr}\left(\rho \frac{\mathbb{1}_d}{d} \text{tr} X\right) = \frac{1}{d} \text{tr}(X) \underbrace{\text{tr}(\rho)}_{=1} = \text{tr}\left(\frac{\mathbb{1}_d}{d} X\right) = \text{tr}(P^*(\rho) X), \quad (2.31)$$

where any input state ρ is mapped to the maximally mixed state $\frac{\mathbb{1}_d}{d}$. In the qubit case this means that the completely depolarizing channel shrinks the Bloch sphere to its origin. If we consider other points of the state space as the image of a CDC, we find that every point defines a suitable channel. This leads us to the whole set of CDCs, which is given by

$$P_\tau^*(\rho) = \tau \quad \forall \rho \in \mathcal{S}(\mathbb{C}^d). \quad (2.32)$$

¹ Abbreviation: CDC

In the Heisenberg picture this is expressed as

$$P_\tau(A) = \text{tr}(\tau A) \mathbb{1}_d \quad \text{with} \quad \tau \in \mathcal{S}(\mathbb{C}^d). \quad (2.33)$$

Considering the corresponding Choi matrix of these channels we can verify complete positivity:

$$\sigma_{P_\tau} = \frac{1}{d} \sum_{i,j} P_\tau(E_{i,j}) \otimes E_{i,j} = \frac{1}{d} \mathbb{1}_d \otimes \bar{\tau} \geq 0. \quad (2.34)$$

In the literature this degree of freedom is mostly neglected and the term of "the" completely depolarizing channel, which preserves the maximally mixed state, is used. We refer to channels, where

$$T^*\left(\frac{\mathbb{1}}{d}\right) = \frac{\mathbb{1}}{d} \quad (2.35)$$

or respectively in the Heisenberg picture

$$\text{tr}[T(X)] = \text{tr} X \quad (2.36)$$

holds, as *bistochastic* channels. Therefore we refer to the CDC, where the maximally mixed state is preserved, as *bistochastic CDC*, keeping in mind that every point of the state space defines another CDC. Furthermore we define a *pure CDC* as a CDC which preserves a pure state. Due to the importance of the bistochastic CDC within this thesis we present the previously used representations of it.

Example 2.6 (Representations of the Bistochastic CDC). First we use (2.34) to compute the corresponding Choi matrix of the bistochastic CDC:

$$\sigma_P = \frac{1}{d} \mathbb{1} \otimes \frac{1}{d} \mathbb{1} = \frac{\mathbb{1}_{d^2}}{d^2}. \quad (2.37)$$

Therefore, the corresponding Choi matrix is the completely mixed state. This illustrates again that the rank of a channel and the rank of its Choi matrix usually do not coincide.

There is a nice way to describe a Kraus decomposition of the bistochastic CDC if we utilize a basic result from representation theory which is known as Schur's lemma [Sim96].

Theorem 2.4 (Schur's Lemma). *Let U_g be an irreducible unitary representation of the group G . Then the following holds:*

$$\exists M : U_g^* M U_g = M \quad \forall g \in G \quad \text{then} \quad M = c_M \mathbb{1}. \quad (2.38)$$

If we choose $T(X) := \frac{1}{|G|} \sum_{g \in G} U_g^* X U_g$ it follows that the theorem can be applied:

$$U_h^* T(X) U_h = \frac{1}{|G|} U_h^* \left(\sum_{g \in G} U_g^* X U_g \right) U_h = \frac{1}{|G|} \sum_{\tilde{g} \in G} U_{\tilde{g}}^* X U_{\tilde{g}} = T(X) \quad \forall h \in G. \quad (2.39)$$

By Schur's lemma it follows that $T(X) = c_X \mathbb{1}$ where c_X can be obtained via:

$$c_X = \frac{1}{d} \text{tr}(T(X)) = \frac{1}{d|G|} \sum_{g \in G} \text{tr}(U_g^* X U_g) = \frac{1}{d} \text{tr} X. \quad (2.40)$$

Therefore we see that the channel $T(X)$ and the bistochastic CDC $P(X)$ are equal. When we consider $P(X)$ on qubits, the Kraus decomposition can be written as a mixture over Pauli channels:

$$P(X) = \frac{1}{4} \sum_{i=0}^3 \sigma_i^* X \sigma_i, \quad (2.41)$$

For a matrix representation of the bistochastic CDC we choose the matrix units as a basis:

$$\tilde{T}_{ij,kl} = \text{tr}(E_{j,i} \frac{1}{d} \text{tr}(E_{k,l}) \mathbb{1}) = \frac{1}{2} \delta_{i,j} \delta_{k,l}. \quad (2.42)$$

Evaluating this expression in the case of qubits we find,

$$\tilde{T} = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}. \quad (2.43)$$

2.5 Roots of Channels

Considering the field of complex numbers, the k -th root of an element $z \in \mathbb{C}$ is defined as another element $w \in \mathbb{C}$, which is equal to z if it is taken to the power of k , i.e. $w^k = z$. The fundamental theorem of algebra ensures the existence and uniqueness of k such elements and there are methods for computing them approximately, even for large k . Applying this approach to quadratic matrices by defining A as the k -th root of B , when $A^k = B$, it is much harder to find a solution or at least to say something about the existence as such. For example the matrix equation

$$A^2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad A \in M_2(\mathbb{C}) \quad (2.44)$$

has no solution. Nevertheless, we define roots for the class of channels with the same input and output space in the following way.

Definition 2.5 (Root of a Channel). A **k -th root** of the channel $T : M_n(\mathbb{C}) \rightarrow M_n(\mathbb{C})$ is a channel $S : M_n(\mathbb{C}) \rightarrow M_n(\mathbb{C})$ with

$$S^r = T \text{ for } r = k \text{ and } S^r \neq T \text{ for } r < k, \quad k, r \in \mathbb{N}. \quad (2.45)$$

We refer to k as the **order** of a root.

Figure 2.6 illustrates the physical interpretation of a root of a channel. The channel S divides the process represented by T into k equal processes.

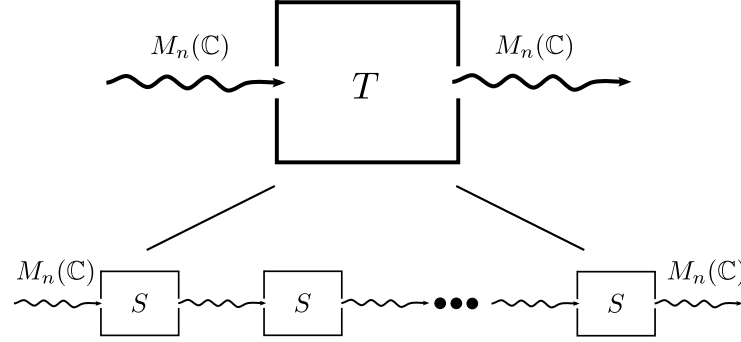


Figure 2.6: If the channel S acts after a certain number of concatenations exactly like the channel T , we refer to S as a root of T . This definition is only reasonable for channels with same input and output system. In a more physical language we say, that the process represented by T can be divided into the process S .

2.5.1 Jordan Decomposition of a Finite CDC-Root

We now examine the problem of finding a root for a CDC as a matrix equation. If we choose the bistochastic CDC with a qubit system as an input and output space and the matrix units as an operator basis we get

$$\tilde{R}^k = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \quad (2.46)$$

where $\tilde{R}$ denotes the representation matrix of a possible root. Solving this equation directly by a parametrization of possible representation matrices for channels and matrix multiplication can be assumed to be impossible (especially for large k or in higher dimensions). But we can give a statement for the maximal order of a finite root. For this purpose we recall some facts of basic linear algebra:

Theorem 2.5 (Jordan Decomposition [Wol09]). *For every $T \in M_d(\mathbb{C})$ there is a similarity matrix S , s.t.*

$$T = S \underbrace{\left(\bigoplus_{k=1}^K J_k(\lambda_k) \right)}_J S^{-1}, \quad J_k(\lambda) := \begin{pmatrix} \lambda & 1 & & \\ & \ddots & \ddots & \\ & & \lambda & 1 \\ & & & \lambda \end{pmatrix} \in M_{d_k}(\mathbb{C}), \quad (2.47)$$

where the matrix J is called the Jordan decomposition of T and the $J_k(\lambda)$ are the Jordan blocks of size d_k corresponding to the eigenvalue λ .

The number of Jordan blocks with the eigenvalue λ is the *geometric multiplicity* while the sum of the dimensions $\sum_{\lambda=\lambda_k} d_k$ is the *algebraic multiplicity*.

The Jordan decomposition of a possible root is the key for obtaining an upper bound for the value of k in (2.46). Since every CDC has $\{1, 0, \dots, 0\}$ as its distribution of eigenvalues

and is a map with rank one, its Jordan decomposition looks like:

$$J_{CDC} = \begin{pmatrix} 1 & 0 \\ 0 & \mathbf{0} \end{pmatrix}. \quad (2.48)$$

Due to the fact that the multiplication of a matrix with itself is for the eigenvalues just a multiplication with themselves, we find that any possible (finite) root needs to have the same fixed distribution of eigenvalues as a CDC. This leads to a finite set of possible Jordan decompositions in theorem 2.5, which consist of a one dimensional Jordan block referring to the eigenvalue one and some nilpotent Jordan blocks according to the geometric multiplicity of the eigenvalue zero. Therefore the set of possible values of the order is bounded:

Theorem 2.6. *Let $T : \mathcal{B}(\mathbb{C}^d) \rightarrow \mathcal{B}(\mathbb{C}^d)$ be a quantum channel, then the highest possible order of a root of T is $d^2 - 1$.*

Proof. Recall that T can be represented by a $d^2 \times d^2$ matrix. Considering the finite set of possibilities for the Jordan decomposition of a root of the CDC and their behavior in the case of concatenation, we see that the nilpotent part of the Jordan decomposition vanishes as late as possible if it consists only of two Jordan blocks:

$$J = \begin{pmatrix} 1 & 0 \\ 0 & J(0) \end{pmatrix} \in M_{d^2}(\mathbb{C}) \quad J(0) := \begin{pmatrix} 0 & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{pmatrix} \in M_{d^2-1}(\mathbb{C}). \quad (2.49)$$

After $d^2 - 1$ steps of concatenation this Jordan decomposition is the same as the decomposition for the CDC. $\square$

Notice that this theorem states neither existence nor uniqueness of a root, which is still left as a question. If we apply this theorem to (2.46), we find that the matrix exponent k is smaller or equal to three.

CHAPTER 3

The Classical Setting

In this chapter we consider *classical stochastic processes* as a kind of pre-stage to the quantum systems. Although classical systems have less degrees of freedom, we hope that we can abstract from the question of maximal roots in the classical case to the quantum case. For this purpose we first put the previously described definitions of quantum systems in the context of classical systems. We find that the question of roots for the classical analogue of completely depolarizing channels can exhaustively be solved in the case of a three dimensional classical system. We furthermore introduce an approach that can be used to construct maximal roots in arbitrary dimensions. Finally we discuss the embedding of maximal roots from a classical into a quantum system.

3.1 Classical Systems

We introduced quantum mechanical systems consisting of states, channels and observables, which describe a complete statistical experiment. We will reformulate these expressions in the context of classical probability theory. Identically to the quantum case, we restrict our description to a finite set of possible outcomes $\mathfrak{X} = \{1, \dots, n\}$ with corresponding probabilities $\sum_{x \in \mathfrak{X}} p(x) = 1$. The set of states can be represented by a finite dimensional vectors with the probabilities as their entries. This set forms a convex set which allows for every point a unique decomposition into extreme points, i.e. a *simplex*. Figure 3.1 illustrates the possible set of states for $\mathfrak{X} = \{1, 2, 3\}$. Any two classical observables should be jointly measurable¹. This is reflected by choosing the $d \times d$ diagonal matrices as the observable algebra. This restriction causes the impossibility of many quantum effects, e.g. the *uncertainty principle*.

¹ In a larger context this can be identified as a definition of a classical system

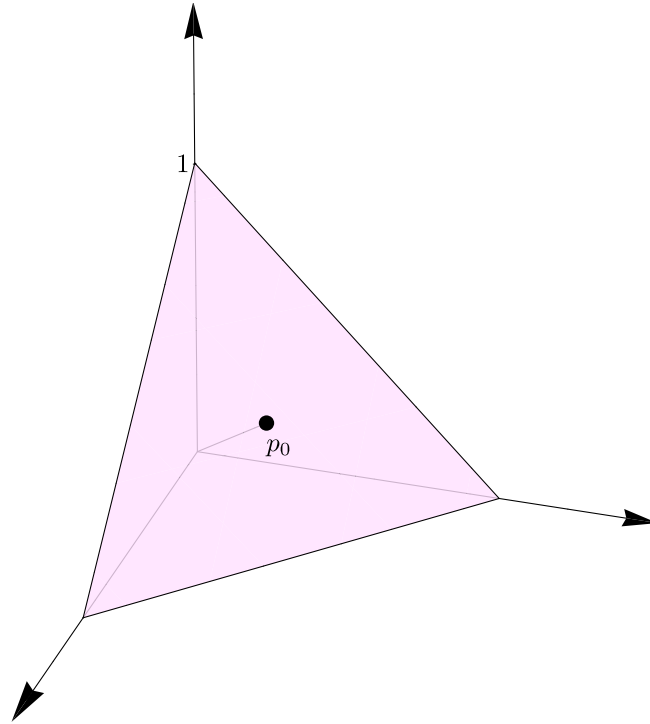


Figure 3.1: The state space of a classical system with 3 possible outcomes. Contrary to the Bloch sphere the state space is a simplex. The point p_0 labels the completely mixed state.

3.1.1 Classical Channels

The classical analogue of a quantum channel is represented by the class of *stochastic* matrices.

Definition 3.1 (Stochastic Matrix). A Matrix $A \in M_d(\mathbb{R})$ is called **stochastic**, if

$$A_{i,j} \geq 0 \quad \text{and} \quad \sum_i A_{i,j} = 1 \quad \forall i,j \in \{1, \dots, d\}. \quad (3.1)$$

By definition of the matrix product we can see that stochastic matrices map classical states to classical states and therefore are suitable for representing classical channels. There are no additional demands for classical channels that must be satisfied, like the complete positivity in the quantum case. If the completely mixed state is preserved under the mapping of a classical channel, it will be represented by a *bistochastic* matrix, i.e. a stochastic matrix A with the additional property $\sum_j A_{i,j} = 1 \quad \forall i \in \{1, \dots, d\}$. It is sometimes useful to illustrate stochastic matrices as a weighted directed graph, where the pure states are the vertices and the entries of the matrix represent the transition probabilities. Figure 3.2 depicts an easy example.

3.1.2 Classical CDCs

To define the set of completely depolarizing channels we recall that in the quantum case this was defined via a channel with rank one such that it shrinks the state space immediately to a point. This leads to the following characterization for classical CDCs:

Lemma 3.1 (Classical CDCs). *A classical channel is completely depolarizing, if and only if all columns of its matrix are equal.*

Proof. Trivially the channel has rank one if all columns are equal. Conversely, a stochastic matrix maps states to states, such that the first column of the matrix (which is the image of a pure state) is a non zero vector. Due to the rank, the remaining columns can only be filled with the same or the null vector. Of course the null vector violates the stochasticity of the matrix. $\square$

We furthermore find that the determining vector of the matrix represents the point the state space is mapped to. If this vector is a unit vector the channel is a pure CDC. Equivalently to the quantum case we refer to the CDC which preserves the completely mixed state as the *bistochastic* CDC. Its matrix and transition graph are illustrated in the following figure 3.2 .

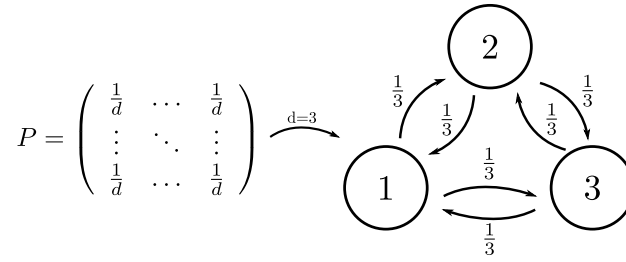


Figure 3.2: The bistochastic CDC preserves the completely mixed state and is represented via the bistochastic matrix with identical entries everywhere. The corresponding transition graph is therefore completely symmetric.

3.1.3 Roots of Classical CDCs

Because classical channels are represented by stochastic matrices, we can simply adapt the definition of a root from the quantum case, i.e. a matrix A is a k -th root of P if $A^k = P$. The upper bound for k can be derived in an analogous way to the quantum case, except that we replace the dimension of the representation matrices from $d^2 \times d^2$ to $d \times d$. Therefore $d - 1$ is the upper bound for the order of a root of a classical CDC. If we consider the Perron-Frobenius theorem [Bre99] for primitive stochastic matrices, i.e. a square stochastic matrix S where there is some $n \in \mathbb{N}$ such that the matrix A^n has no entries equal to zero, we get that the distribution of eigenvalues of a primitive stochastic matrix is given by $\{1, \lambda_1, \dots, \lambda_n\}$ whereby $1 < |\lambda_1| \leq |\lambda_2| \leq \dots \leq |\lambda_n|$.¹ This means that

¹ The constraint of being primitive is not so important for our purpose, because every root fulfils it.

every primitive stochastic matrix converges to a CDC, if we self concatenate it infinitely often. The question is how this convergence can be obtained in a *finite* number of steps.

3.2 Pure CDCs

To construct roots of the highest possible order for a pure CDC, we first consider its matrix and its directed graph. An example is depicted in figure 3.3. The transition graph

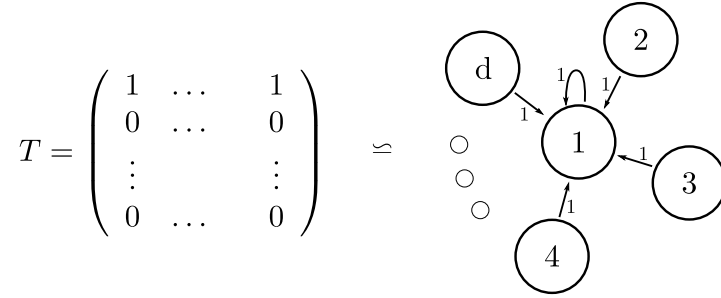


Figure 3.3: The directed graph of a pure CDC. Every state will be mapped with probability one to a pure state.

of any other pure classical CDC follows simply by permutation of the states. We notice that every state is surely mapped to the preserved state. Considering the behavior of the concatenation in figure 3.4, we find that roots of the pure CDC can be constructed by building paths inside the transition graphs, i.e. first to map a state with certainty to another than the preserved one. Obviously, the length k of the path, which is equal to the states involved, is the order of the root. If we concatenate the channel with itself we see that in every step of concatenation the length of the path is lowered by one, s.t. after $k - 1$ steps it is identical to the CDC. Moreover, if we consider the matrix representation

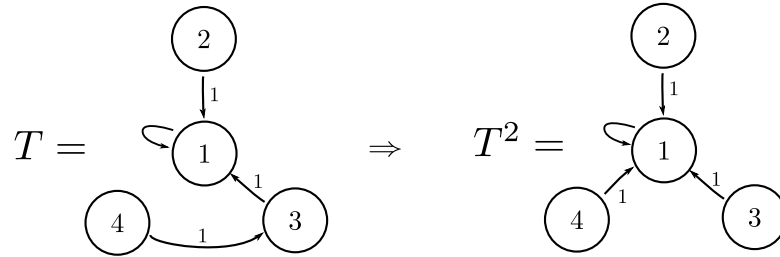


Figure 3.4: This depicts a root of second order of a pure CDC for a four dimensional setting. We find that the graph of T contains a path from 4 to 3 to 1. If we extend this path, the order of the root increases.

of a maximal path, we find that a maximal root of the pure CDC can be expressed in

matrix notation as:

$$T_{max} = \begin{pmatrix} 1 & 1 & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (3.2)$$

This construction is completely independent from the underlying labeling of the states.

3.3 The Bistochastic CDC

To construct roots of the bistochastic CDC it is useful to perform some further investigation of bistochastic matrices. One remarkable result on bistochastic matrices is the following theorem, which we use to discuss a special property of a root.

Theorem 3.1 (Birkhoff [Bir46]). *Let $S \in M_n(\mathbb{R}_+)$ be a bistochastic matrix. Then it can be written as a convex combination of permutation matrices P_i , i.e.*

$$S = \sum_i p_i P_i \quad \text{with} \quad \sum_i p_i = 1. \quad (3.3)$$

A possible decomposition into permutations of the bistochastic CDC is

$$S_{CDC} = \sum_{P_i \in \mathfrak{S}_n} \frac{1}{n!} P_i, \quad (3.4)$$

where $\mathfrak{S}_n$ denotes the symmetric group.

3.3.1 Roots in 3D

Since the maximal order of a classical root is determined as $d - 1$ with respect to its dimension d , a classical system needs to be three dimensional for the existence of non-trivial roots. In this case we can find a graphical representation of the problem due to its constraints. First we assume that a possible root of the bistochastic CDC also needs to be bistochastic. In order for T being root, we see that the distribution of its eigenvalues $\lambda_i \in \{1, 0, 0\}$ implies $\text{tr } T = 1$. If we evaluate these constraints we find a parametrization of possible roots in the following way:

$$T = \begin{pmatrix} a_1 & 2 - 2a_1 - a_2 - 2a_3 & -1 + a_1 + a_2 + 2a_3 \\ a_2 & a_3 & 1 - a_2 - a_3 \\ 1 - a_1 - a_2 & -1 + 2a_1 + a_2 + a_3 & 1 - a_1 - a_3 \end{pmatrix} \quad a_i \in [0, 1]. \quad (3.5)$$

The nonnegativity of the remaining entries of T leads to a convex polytope as an intersection between six half-spaces. This can be understood as the set of possible configurations for $\{a_1, a_2, a_3\}$ and is depicted in figure 3.5. The points on the surface of the polytope represent matrices where one of the coefficients is zero. The eigenvalues furthermore imply that for a possible root the determinant vanishes. This condition forms a quadric which is illustrated in the same figure.

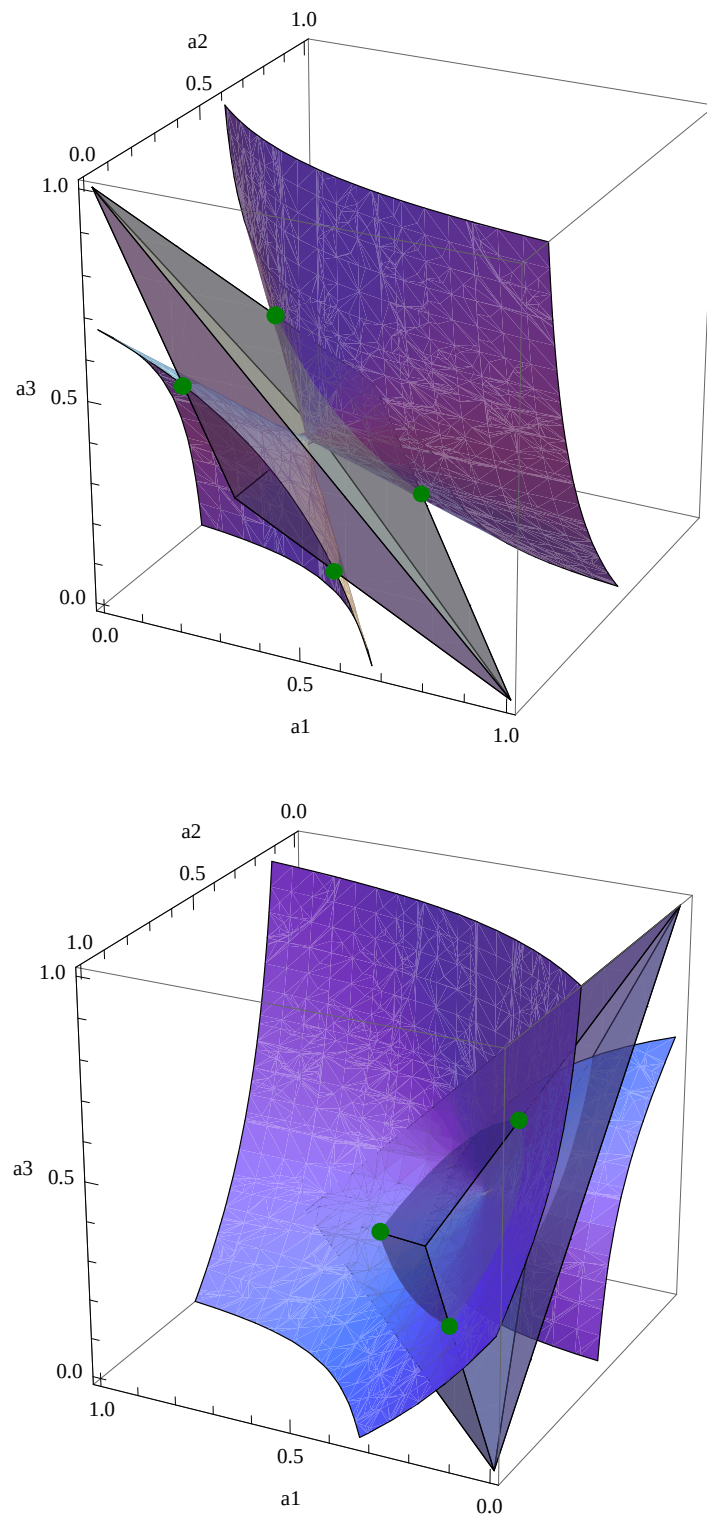


Figure 3.5: This figure is a plot of the state space against the parameters a_1, a_2, a_3 from (3.5). The polyhedron marks the set of possible configurations for three dimensional bistochastic matrices with trace one. If we look at the intersection between the area where $\det T = 0$ and this convex subset of the unit cube, we get all possible square roots of the bistochastic CDC. The configuration of the CDC lies at the touching point of the two hyperboloids. The green spots mark the configuration of a root which is the convex combination of three permutations.

If we look at the intersection of both we get the possible configurations for roots of the bistochastic CDC, where the point $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ represents the bistochastic CDC itself.

Since we are considering a three dimensional system the other points just represent roots of order two.

Let us consider the possible roots in connection to Birkhoff's theorem. One may ask if there is a least number of permutations needed if we decompose the possible roots into convex combinations. Roughly speaking, the number of used permutations corresponds to the non-zero entries of a bistochastic matrix, which means we are looking for a root with as many zero entries as possible. If we take a closer look at the intersection area we find that there are only six points which lie on a face of the polytope and therefore represent matrices with the maximal number of two zero entries. Exemplarily we discuss the root represented by the point $\{a_1 = \frac{1}{6}(3 - \sqrt{3}), a_2 = 0, a_3 = \frac{1}{\sqrt{3}}\}$:

$$T = \begin{pmatrix} \frac{1}{6}(3 - \sqrt{3}) & 1 - \frac{1}{\sqrt{3}} & \frac{1}{2}(-1 + \sqrt{3}) \\ 0 & \frac{1}{\sqrt{3}} & 1 - \frac{1}{\sqrt{3}} \\ \frac{1}{6}(3 + \sqrt{3}) & 0 & \frac{1}{6}(3 - \sqrt{3}) \end{pmatrix}. \quad (3.6)$$

This can be written as the convex combination of three permutations:

$$T = \frac{1}{2}(-1 + \sqrt{3}) P_{321} + \frac{1}{6}(3 - \sqrt{3}) P_{123} + 1 - \frac{1}{\sqrt{3}} P_{312}. \quad (3.7)$$

If one considers the other five roots it turns out that in every case three permutations are necessary, such that this answers the question about the minimal number of permutations for a root in the three dimensional case.

3.3.2 Roots in Higher Dimensions

The number of parameters to describe the set of bistochastic matrices in a d -dimensional system is $(d - 1)^2$. Even if we regard the constraints resulting from the desired eigenvalue distribution, a graphical solution becomes impossible for dimensions larger than three. Therefore we need to choose a new approach to construct roots of larger order.

Since we know the Jordan decomposition of a root, we can get some information about the properties of the Jordan basis. Before we do this let us first clear up some notation:

Let $B = \{e_1, \dots, e_d\}$ be a basis of a d -dimensional vector space V . Then its *dual* basis $B^* = \{e^1, \dots, e^d\}$ is uniquely defined via $e^i(e_j) = \delta_{i,j}$ and is a basis of the dual vector space V^* . In this context we can write elements of V as column vectors and elements of V^* as row vectors that act as linear functionals by left matrix multiplication. With this notation $e^i e_j$ is a number and $e_j e^i$ denotes a matrix. If we consider a matrix A as a linear map on V , the transformation of the entries by changing the basis B to $\tilde{B}$ is described by:

$$\tilde{A} = T A T^{-1} = \begin{pmatrix} | & & | \\ \tilde{e}_1 & \dots & \tilde{e}_n \\ | & & | \end{pmatrix} A \begin{pmatrix} - & \tilde{e}^1 & - \\ & \vdots & \\ - & \tilde{e}^n & - \end{pmatrix} = \sum_{i,j} a_{i,j} \tilde{e}_i \tilde{e}^j, \quad (3.8)$$

where $a_{i,j}$ denote the coefficients of A .

Thus we write the Jordan decomposition of a root of highest order T as follows,

$$T = e_1 e^1 + \sum_{i=2}^{d-1} e_i e^{i+1}, \quad (3.9)$$

for any Jordan base $B_J = \{e_1, \dots, e_d\}$ and dimension d . Now we consider T^{d-1} which needs to be the bistochastic CDC:

$$T^{d-1} = e_1 e^1 \stackrel{!}{=} \frac{1}{d} \begin{pmatrix} 1 & \dots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \dots & 1 \end{pmatrix}. \quad (3.10)$$

Thus we need to choose $e_1 = c_1(1, \dots, 1)$ and $e^1 = c^1(1, \dots, 1)$ where $c_1 c^1 \stackrel{!}{=} \frac{1}{d}$. Without loss of generality we can choose $c_1 = 1$ and $c^1 = \frac{1}{d}$. Since e_1 is the corresponding eigenvector to the eigenvalue one, we find $T e_1 = e_1$ and $e_1^\top T = e_1^\top$ due to the form of the Jordan decomposition. These are just compact notations for $\sum_j T_{i,j} = 1$ and $\sum_i T_{i,j} = 1$. From this follows, that every possible root of the bistochastic CDC needs to be bistochastic itself. Furthermore, we find by definition of the dual basis that the sum of the components of every but the first vector of the basis is zero, i.e. $\sum_k (e_i)_k = 0$ and $\sum_k (e^i)_k = 0$ for $i \in \{2, \dots, d\}$. If we consider (3.9) we see that choosing the elements $\{e_2, \dots, e_d\}$ randomly would violate the positivity of T . Therefore we restrict the construction with the Jordan decomposition in two ways. First we choose the elements $\{e_2, \dots, e_d\}$ to be real such that we can a priori exclude complex entries of T and secondly we weight the Jordan block corresponding to the eigenvalues 0 with ε . This leads to the following theorem:

Theorem 3.2 (Maximal Bistochastic Root). *Let $B := \{e_1 = (1, \dots, 1), e_2, \dots, e_d\}$ be a basis of the real vector space V and $B^* := \{e^1 = (\frac{1}{d}, \dots, \frac{1}{d}), \dots, e^d\}$ its dual basis. Then there is an $\varepsilon > 0$, such that*

$$T = e_1 e^1 + \varepsilon \sum_{\alpha=2}^{d-1} e_\alpha e^{\alpha+1} = \begin{pmatrix} \frac{1}{d} & \dots & \frac{1}{d} \\ \vdots & \ddots & \vdots \\ \frac{1}{d} & \dots & \frac{1}{d} \end{pmatrix} + \varepsilon \sum_{\alpha=2}^{d-1} e_\alpha e^{\alpha+1} \quad (3.11)$$

is a $(d-1)$ -th root of the bistochastic CDC.

Proof. The properties $\sum_j T_{i,j} = 1$ and $\sum_i T_{i,j} = 1$ are already assured by choosing the element $e_1 = (1, \dots, 1)$. Furthermore the multiplication with ε does not change the behavior under concatenation. What is still left to show is that all entries of T are nonnegative for some ε . This is expressed by the following constraint:

$$T_{i,j} \geq 0 \quad \Leftrightarrow \quad \left| \left(\varepsilon \sum_{\alpha=2}^{d-1} e_\alpha e^{\alpha+1} \right)_{i,j} \right| \leq \frac{1}{d}. \quad (3.12)$$

1 Here $e_1^\top$ denotes the transposed vector.

From this we derive the ε as follows:

$$\begin{aligned} \left| \varepsilon \left(\sum_{\alpha=2}^{d-1} e_{\alpha} e^{\alpha+1} \right)_{i,j} \right| &\leq \varepsilon \sum_{\alpha=2}^{d-1} \left| (e_{\alpha} e^{\alpha+1})_{i,j} \right| \leq \varepsilon \sum_{\alpha=2}^{d-1} |e_{\alpha}|_{\infty} |e^{\alpha+1}|_{\infty} \\ &\leq \varepsilon (d-2) \max_{\alpha \in \{2, \dots, d-1\}} \left\{ |e_{\alpha}|_{\infty} |e^{\alpha+1}|_{\infty} \right\} \leq \frac{1}{d}, \end{aligned}$$

where we choose

$$\varepsilon := \frac{1}{d(d-2) \max_{\alpha \in \{2, \dots, d-1\}} \{|e_{\alpha}|_{\infty} |e^{\alpha+1}|_{\infty}\}}. \quad (3.13)$$

The ε is always well-defined since the $\{e_{\alpha}\}$ form a basis and therefore do not contain the null vector. $\square$

After we have shown the existence of a root of highest order we can again consider the connection to Birkhoff's theorem and ask about the number of necessary permutations. For this purpose it is useful to regard the entries of T as a continuous function of ε . If we choose $\varepsilon = 0$ all entries are equal to $\frac{1}{d}$. Since $\sum_k (e_i)_k = 0$ and $\sum_k (e^i)_k = 0$ for $i \in \{2, \dots, d\}$ we will decrease some entries of T , if we increase ε continuously. Finally we will reach a ε where at least one entry is equal to zero. This means that we can construct a root with a convex decomposition into $(d-1)!$ permutations.

3.4 Embedding into Quantum Systems

As we have seen, the set of classical systems is a subset of the quantum systems. For example the state space of a two dimensional classical system, a classical bit, can be embedded into the Bloch sphere as the line between two antipodal points. In the following we will explore the relationships between classical and quantum channels, that are presented in [WPG10].

If we consider the representation matrix of a quantum channel T in the basis of matrix units ($\tilde{T}_{ij,kl} = \langle j | T(E_{k,l}) | i \rangle$) we find that $\tilde{T}_{ii,jj}$ is a nonnegative entry due to the positivity of T and $\sum_j \tilde{T}_{ii,jj} = 1$ due to its unitality. Therefore every representation matrix of a quantum channel contains a stochastic matrix $S_{i,j} := \tilde{T}_{ii,jj}$ as a principal matrix. If in addition the completely mixed state is preserved, the principal matrix is bistochastic. To construct a representation of the quantum channel T from a stochastic matrix S we consider the following map $T^* : \mathcal{S}(\mathbb{C}^d) \rightarrow \mathcal{S}(\mathbb{C}^d)$,

$$T^*(\rho) := \sum_{i,j=1}^d S_{i,j} \langle j | \rho | j \rangle | i \rangle \langle i |. \quad (3.14)$$

In order to check complete positivity of that map we consider the corresponding Choi matrix:

$$\sigma_{T^*} = \frac{1}{d} \sum_{k,l} T^*(E_{k,l}) \otimes E_{k,l} = \frac{1}{d} \sum_{k,l} S_{k,l} |k\rangle \langle k| \otimes |l\rangle \langle l|. \quad (3.15)$$

Since this is a diagonal matrix with the entries of S (divided by d) as eigenvalues, it is positive and the map T^* is completely positive. The representation matrix of this map in

the basis of matrix units is given by:

$$\tilde{T}^*_{mn,kl} = \langle n | T^*(E_{k,l}) | m \rangle = \langle n | \left(\sum_{i,j=1}^d S_{i,j} |j\rangle \langle E_{k,l}|j\rangle \langle i| \right) | m \rangle = S_{n,k} \delta_{l,k} \delta_{n,m}. \quad (3.16)$$

This matrix consists of S as principal matrix and the rest of the entries are zero. The map T^* can therefore be understood as the embedding of a classical channel into a quantum channel.

We furthermore can perform an embedding of a maximal classical root S into a quantum channel $\tilde{S}$ via (3.14). Since the entries of S then determine the eigenvalues of the corresponding Choi matrix $\sigma_{\tilde{S}}$, the number of zero entries determines the Kraus rank. The order of the root is then still the same, but it is obviously not maximal anymore, since the upper bound for a root in the quantum case is $d^2 - 1$.

CHAPTER 4

The Qubit Roots

After solving the problem of constructing roots of maximal order in the classical case, we consider the most elementary quantum case, where we restrict the input and output system of a quantum channel to $\mathcal{H} = \mathbb{C}^2$. Considering theorem 2.6 we find that this is the smallest possible dimension of non-trivial roots.

In this chapter we first take a look at reversible channels $T(X) = U^* X U$. Then we derive that every bistochastic qubit channel can be written as a composition of a unitary and a Pauli diagonal channel, i.e. a channel where $T(\sigma_i) = \lambda_i \sigma_i$. Furthermore, we discuss which compositions represent a root of maximal order. Finally we discuss, how these qubit roots can be used to construct roots in higher dimensions. In what follows, every representation matrix of a channel will be given in the basis of Pauli operators $\{\mathbf{1}, \sigma_1, \sigma_2, \sigma_3\}$, unless something else is stated.

4.1 Unitary Qubit Channels

Let us first consider channels with Kraus rank one. This leads us to the set of *unitary* qubit channels, i.e.

$$T^*(X) = U X U^* \quad U \in U(2). \quad (4.1)$$

This set of channels represents the reversible channels. Moreover, we find that these channels preserve the completely mixed state and hence are bistochastic. Before we characterize the action of these channels on the state space, we take a closer look at the unitary matrices U in two dimensions. If we evaluate the constraints of unitarity, $U^{-1} = U^*$ which implies $|\det U| = 1$, we can parameterize an element via:

$$U = e^{i\phi} \begin{pmatrix} u_0 + iu_3 & u_1 + iu_2 \\ -u_1 + iu_2 & u_0 - iu_3 \end{pmatrix} \quad \text{with } u_i \in \mathbb{R} \quad \text{and} \quad \sum_i u_i^2 = 1. \quad (4.2)$$

Since we consider this parametrization in the context of channels from (4.1), we can neglect the parameter ϕ and regard just elements from $SU(2)$. This group is topologically isomorphic to the 3-sphere (the unit sphere in $\mathbb{R}^4$). The following lemma arises due to the connection between the $SU(2)$ and the $SO(3)$ group and can be found in [Stu64].

Lemma 4.1. *Every unitary qubit channel $T_U^*(X) = UXU^*$ corresponds to the following representation matrix:*

$$\widetilde{T}_U^* = \begin{pmatrix} 1 & 0 \\ 0 & \mathbf{R}_U \end{pmatrix}, \quad (4.3)$$

where R_U is an element of $SO(3)$. Therefore a unitary channel is a rotation of the Bloch sphere around its origin.

Proof. If one considers the representation matrices of the channels induced by unitaries, parameterized as in (4.2), one obtains the following:

$$\widetilde{T}_U^* = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & u_0^2 + u_1^2 - u_2^2 - u_3^2 & 2(u_1u_2 + u_0u_3) & -2u_0u_2 + 2u_1u_3 \\ 0 & 2u_1u_2 - 2u_0u_3 & u_0^2 - u_1^2 + u_2^2 - u_3^2 & 2(u_0u_1 + u_2u_3) \\ 0 & 2(u_0u_2 + u_1u_3) & -2u_0u_1 + 2u_2u_3 & u_0^2 - u_1^2 - u_2^2 + u_3^2 \end{pmatrix}. \quad (4.4)$$

The 3×3 submatrix R_U of $\widetilde{T}^*$ with polynomial entries of u_i consists of orthogonal columns and its determinant is equal to one. Hence, the R_U form a subgroup of $SO(3)$. If we consider the map between $SU(2)$ and its corresponding submatrix R_U of $\widetilde{T}^*$ we see that it is continuous. Since $SU(2)$ is compact and connected, we conclude that the group formed by the R_U is also compact and connected. The only compact connected subgroups of $SO(3)$ are the group of rotations about a fixed axis, the identity and $SO(3)$ itself. Due to the non-zero entries of R_U , it leaves no axis fixed and it follows $R_U = SO(3)$. Note that $R_U = R_V$ if and only if $U = -V$ or $U = V$, such that this is a two-to-one map. $\square$

If we use the previously introduced notation of a qubit state as a point of the Bloch sphere $\rho(\vec{r}) = \frac{1}{2}(\mathbb{1} + \vec{r}\vec{\sigma})$, we see that every unitary channel $T^*(\rho) = U\rho U^*$ can be written as $T^*(\rho(\vec{r})) = \frac{1}{2}(\mathbb{1} + (R_U\vec{r})\vec{\sigma})$.

4.2 Bistochastic Qubit Channels

After we have discussed unitary qubit channels, we now consider more general channels. In fact, we will derive that an arbitrary bistochastic qubit channel can be obtained by a composition of unitary and Pauli diagonal channels, i.e. a channel which maps Pauli matrices to themselves. We consider the set of possible channels related to [RSW02, KR01, HW07]. Like in the classical case, a root of the bistochastic CDC needs to be bistochastic itself. The following theorem and its proof can be found in [KR01]:

Theorem 4.1 (Decomposition of Qubit Channels). *Every bistochastic qubit channel can be decomposed as*

$$T^*(\rho) = U_2 \Lambda [U_1 \rho U_1^*] U_2^*, \quad (4.5)$$

where U_1 and U_2 are unitaries and Λ is a Pauli diagonal map, i.e. $\Lambda[\sigma_i] = \lambda_i \sigma_i$, $i = 1, 2, 3$.

Proof. If we consider the set of matrices $\widetilde{T}^*$ of linear maps on the Bloch sphere, we find that it is trace-preserving if and only if the first row satisfies $\widetilde{T}_{1,k}^* = \delta_{1,k}$, i.e.

$$\widetilde{T}^* = \begin{pmatrix} 1 & 0 \\ \mathbf{m} & \mathbf{M} \end{pmatrix}. \quad (4.6)$$

The map is bistochastic if and only if $\mathbf{m} = 0$. We see that the submatrix M needs to consist of real entries, since T^* is an endomorphism on the real vector space of hermitian matrices. If we use the notation $\rho(\vec{r}) = \frac{1}{2}(\mathbb{1} + \vec{r}\vec{\sigma})$ for states we can write:

$$T^*(\rho(\vec{r})) = \frac{1}{2}(\mathbb{1} + (\mathbf{M}\vec{r})\vec{\sigma}), \quad (4.7)$$

with M as a real 3×3 -Matrix. The theorem of the *polar decomposition* states that an arbitrary operator A can be factorized into a product of a unitary and a positive operator $A = UP$. This theorem can also be applied to real matrices, where a real matrix B is then decomposable into an orthogonal and a symmetric matrix, i.e. $B = OS$. If we apply this to the matrix M , we find that $M = O_1P$ where $O_1 \in O(3)$ and $P = P^\top$. Furthermore we diagonalize P by an orthogonal transformation:

$$M = O_1P = \underbrace{O_1O_2}_{R_1}LO_2^* = R_1LR_2 \quad \text{with } O_i \in O(3), L = \text{diag}\{\lambda_1, \lambda_2, \lambda_3\}, R_i \in SO(3), \quad (4.8)$$

where we can ensure $\det R_1 = \det R_2 = 1$ by changing the signs of the λ_i . If we evaluate this in (4.7), we see

$$T^*(\rho(\vec{r})) = \frac{1}{2}(\mathbb{1} + (\mathbf{R}_1\mathbf{L}\mathbf{R}_2\vec{r})\vec{\sigma}). \quad (4.9)$$

The theorem follows from the fact that every rotation of the Bloch sphere corresponds to a unitary channel (see Lemma 4.1). $\square$

Since we have completely characterized the class of unitary channels, we now focus our attention to the Pauli diagonal channels.

4.2.1 Pauli Diagonal Channels

The bistochastic CDC is a simple example of a Pauli diagonal channel. If we recall its Kraus decomposition in the qubit case (see (2.41)), we find $P(X) = \sum_{i=0}^3 \sigma_i^* X \sigma_i$. An interesting classification of all Pauli diagonal channels arises if we allow different weights μ_i for the single Kraus operators:

Theorem 4.2 (Pauli Diagonal Channels [HW07]). *A channel is Pauli diagonal if and only if it admits a Kraus decomposition of the form*

$$T(X) = \sum_{i=0}^3 \mu_i \sigma_i^* X \sigma_i, \quad (4.10)$$

with nonnegative μ_i and $\sum_{i=0}^3 \mu_i = 1$.

Proof. First, let T be a map of the form (4.10). A short calculation shows that it is in fact Pauli diagonal:

$$T(\sigma_0) = (\mu_1 + \mu_2 + \mu_3 + \mu_4)\sigma_0 = \lambda_0\sigma_0, \quad (4.11)$$

and for $j \in \{1, 2, 3\}$:

$$\begin{aligned} T(\sigma_j) &= \sum_{i=0}^3 \mu_i \sigma_i^* \sigma_j \sigma_i = \mu_0 \sigma_j + \sum_{i=1}^3 \mu_i \sigma_i^* \sigma_j \sigma_i = \mu_0 \sigma_j + \sum_{i=1}^3 \mu_i \sigma_i^* (\{\sigma_j, \sigma_i\} - \sigma_i \sigma_j) \\ &= \mu_0 \sigma_j + \sum_{i=1}^3 \mu_i \sigma_i^* (2\delta_{i,j} - \sigma_i \sigma_j) = (\mu_0 + \mu_j - \sum_{i \neq j} \mu_i) \sigma_j = \lambda_j \sigma_j. \end{aligned}$$

□

From this follows the relation between the eigenvalues λ_i and the Kraus weights μ_i :

$$\begin{aligned} \lambda_0 &= \mu_0 + \mu_1 + \mu_2 + \mu_3 = 1 & \lambda_1 &= \mu_0 + \mu_1 - \mu_2 - \mu_3 \\ \lambda_2 &= \mu_0 - \mu_1 + \mu_2 - \mu_3 & \lambda_3 &= \mu_0 - \mu_1 - \mu_2 + \mu_3 \end{aligned} \quad (4.12)$$

Unitality of T imposes $\sum_{i=0}^3 \mu_i = 1$. The nonnegativity of the μ_i follows, if we assume T to be a Pauli diagonal channel, i.e. $S(\sigma_i) = \lambda_i \sigma_i$. The corresponding Choi matrix is given via:

$$\sigma_T = \begin{pmatrix} \frac{\lambda_0 + \lambda_3}{4} & 0 & 0 & \frac{\lambda_1 + \lambda_2}{4} \\ 0 & \frac{\lambda_0 - \lambda_3}{4} & \frac{\lambda_1 - \lambda_2}{4} & 0 \\ 0 & \frac{\lambda_1 - \lambda_2}{4} & \frac{\lambda_0 - \lambda_3}{4} & 0 \\ \frac{\lambda_1 + \lambda_2}{4} & 0 & 0 & \frac{\lambda_0 + \lambda_3}{4} \end{pmatrix}, \quad (4.13)$$

with eigenvalues:

$$\begin{aligned} e_0 &= \frac{1}{4}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \\ e_1 &= \frac{1}{4}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) \\ e_2 &= \frac{1}{4}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \\ e_3 &= \frac{1}{4}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3). \end{aligned}$$

If we consider (4.12), we find that the eigenvalues are equal to the weights in (4.10), i.e. $e_i = \mu_i$, from which the nonnegativity of the μ_i follows. If we recall the proof of theorem 2.1 we see that the Kraus operators $\{K_i\}$ can be constructed out of the spectral decomposition of σ_T . If we apply this to the matrix (4.13) and use the unitary freedom of the Kraus decomposition, we find that T is exactly of the form $T(X) = \sum_i \mu_i \sigma_i^* X \sigma_i$.

The suitable μ_i form a tetrahedron which is depicted in figure 4.1. The set of possible μ_i , where one eigenvalue of the channel vanishes, forms three intersecting squares inside the tetrahedron which are also depicted in figure 4.1. This figure shows that a suitable configuration of μ_i , which assures the choice of eigenvalues of the pancake map (2.17) $\{\lambda_0 = 1, \lambda_1 = 0, \lambda_2 = 1, \lambda_3 = 1\}$ lies not inside the tetrahedron, and hence the map cannot be completely positive.

Since the composition of Pauli diagonal channels with unitary channels will neither affect its rank as linear map nor the Kraus rank, we can gain some information about the relationship between the rank of all bistochastic qubit channels and its number of Kraus

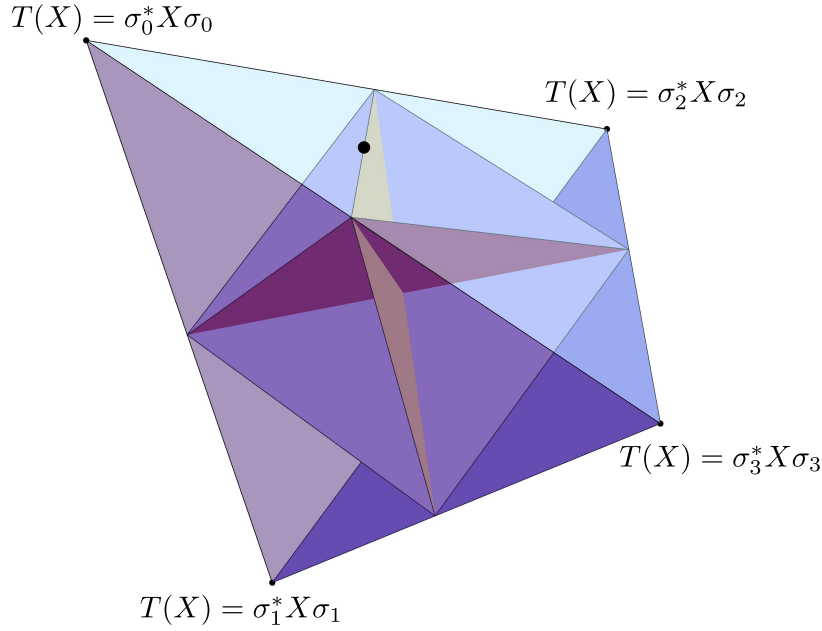


Figure 4.1: The set of possible configurations of the μ_i in equation (4.10) forms a tetrahedron. The extremal points of the tetrahedron represent the configuration where one of the μ_i is equal to one. The number of Kraus operators increases from one to four depending whether the point which characterizes our channel is an extremal point, lies on a line between these points, lies on the surface or inside the tetrahedron. The squares inside the tetrahedron mark the configurations, where one of the eigenvalues of the channel is equal to zero, thus the point in the middle of the tetrahedron, where all squares intersect, represents the bistochastic CDC. The black spot marks the configuration for a channel with rank three and three Kraus operators.

operators, if we consider the intersection of these squares. The number of Kraus operators corresponds to the dimension of the face on which the channel-representing point lies. The extremal points represent channels with one Kraus operator and therefore full rank. If we consider the lines between the extreme points and the points on the surface, we get that a channel with two Kraus operators can exclusively be of rank four or two, since either the point lies on none of the squares or on the intersection of two squares. Conversely channels with three Kraus operators, which are represented by points on the surface of the tetrahedron, can be of rank four or three, since they can lie at most on one square at the same time. Points in the interior of the tetrahedron represent channels with four Kraus operators and can have an arbitrary rank.

4.3 Maximal Qubit Roots

Now that we have discussed the properties of bistochastic qubit channels, let us consider how we can use the knowledge to construct a root of maximal order. Recalling the Jordan

decomposition of a maximal root we find that it is a channel with one dimensional kernel, whereas the eigenvalue zero has the maximal algebraic multiplicity. According to theorem 4.5, we try to construct the desired Jordan decomposition of a root as composition of a Pauli diagonal map, which determines the kernel of the channel, and a suitable rotation, which should increase the algebraic multiplicity of the eigenvalue zero. Since we are looking for a root of maximal order, we need to choose a Pauli diagonal map with rank three. We noticed in the previous section that for this purpose we need a channel with at least three Kraus operators. In the following we discuss exemplarily a construction, which illustrates that this construction is indeed possible.

Example 4.1 (Maximal Qubit Root). Let us consider the channel given by the point $\{\mu_0 = \frac{1}{2}, \mu_1 = 0, \mu_2 = \frac{1}{4}, \mu_3 = \frac{1}{4}\}$ (the black spot in figure 4.1). Its representation matrix is

$$\widetilde{T}_D^* = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} \end{pmatrix}. \quad (4.14)$$

The class of suitable rotation channels can be identified by composing this channel with a parameterized unitary channel $\widetilde{T}_U^*$ (see (4.4)) and subsequent calculation of the eigenvalues of $\widetilde{T}_D^* \widetilde{T}_U^*$:

$$\lambda_0 = 1 \quad \lambda_2 = \frac{1}{2} \left(u_0^2 - u_1^2 - \sqrt{-4u_0^2 u_1^2 + (u_2^2 + u_3^2)^2} \right) \quad (4.15)$$

$$\lambda_1 = 0 \quad \lambda_3 = \frac{1}{2} \left(u_0^2 - u_1^2 + \sqrt{-4u_0^2 u_1^2 + (u_2^2 + u_3^2)^2} \right) \quad (4.16)$$

Since the composition with a unitary channel leaves the geometric multiplicity of the eigenvalue zero unchanged, it will be sufficient to calculate the u_i where λ_2 and λ_3 vanish in order to get the desired Jordan decomposition of a maximal root. For this purpose we take a closer look at λ_2 and λ_3 :

$$\lambda_2 = \lambda_3 = 0 \quad \Leftrightarrow \quad u_0^2 - u_1^2 = 0 \quad \text{and} \quad -4u_0^2 u_1^2 + (u_2^2 + u_3^2)^2 = 0. \quad (4.17)$$

If we incorporate the constraint $\sum_i u_i = 1$, we can find a graphical solution of that problem depicted in figure 4.2. It turns out that the set of possible solutions is the set of points

$$u_0 = \pm \frac{1}{2}, u_1 = \pm \frac{1}{2}, u_2 \in \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right], u_3 = \pm \sqrt{1 - u_2^2}. \quad (4.18)$$

An example of a suitable configuration for the shifter is $\{u_i = \frac{1}{2} | i \in \{1, \dots, 4\}\}$ which characterizes the root:

$$\widetilde{T}^* = \widetilde{T}_D^* \widetilde{T}_S^* = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (4.19)$$

This represents a root of maximal order with the three Kraus operators. But we noticed

that this composition of channels is highly non unique, since every point which lies on only one plane in figure 4.1 together with a suitable rotation defines a possible root.

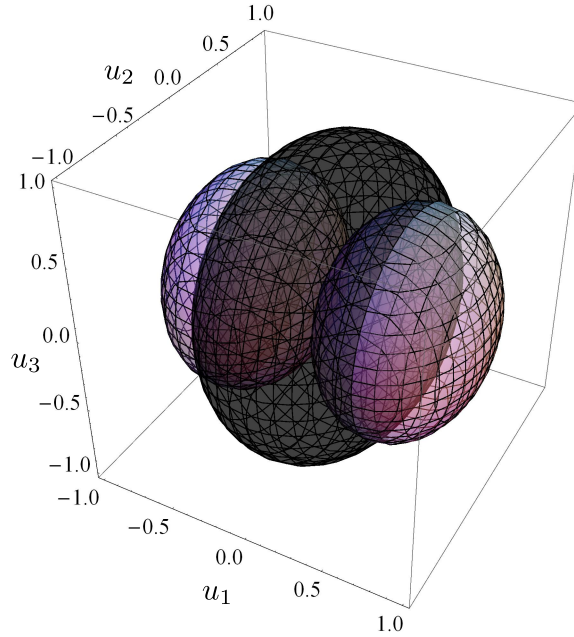


Figure 4.2: The intersection of both areas represent the configurations of u_i where λ_2 and λ_3 in (4.17) are zero.

4.4 Embedding of Qubit Roots into Higher Dimensions

Since we identified the possibility to construct maximal qubit roots, the question arises if these roots could be used to construct roots in higher dimensions. Of course, the considered space needs to be composed of n qubit systems such that the possible dimensions are either $d = 2^n$ for the tensor product of qubit systems or $d = 2n$ for the sum of qubit algebras.

4.4.1 Product Spaces

We start with a construction in the case of a chain of n qubits. The underlying system is then $\bigotimes_{i=1}^n \mathbb{C}_i^2 = \mathbb{C}_1^2 \otimes \mathbb{C}_2^2 \otimes \dots \otimes \mathbb{C}_n^2$ where i denotes the place of a qubit on the chain. The corresponding observable algebras are $\mathcal{B}(\mathbb{C}_1^2) \otimes \mathcal{B}(\mathbb{C}_2^2) \otimes \dots \otimes \mathcal{B}(\mathbb{C}_n^2)$. The idea of embedding a qubit root into that space is to compose a new root out of two channels. First a qubit root which depolarizes on only one system and a swap channel which causes a cyclic permutation of the the algebras on the spin chain. We define the partially depolarizing channel in the following canonical form:

$$T : \bigotimes_{i=1}^n \mathcal{B}(\mathbb{C}_i^2) \rightarrow \bigotimes_{i=1}^n \mathcal{B}(\mathbb{C}_i^2), \quad T(X_1 \otimes \dots \otimes X_n) = P(X_1) \otimes id_{2^{n-1}}(X_2 \otimes \dots \otimes X_n), \quad (4.20)$$

where $P(X)$ denotes any maximal qubit root. Since the permutation of the qubit states is a reversible operation, we can define a unitary permutation operator as follows:

$$U_\pi(\phi_1 \otimes \dots \otimes \phi_n) = \phi_{\pi^{-1}(1)} \otimes \dots \otimes \phi_{\pi^{-1}(n)} \quad \text{for } \phi_1 \otimes \dots \otimes \phi_n \in \bigotimes_{i=1}^n \mathbb{C}_i^2, \quad (4.21)$$

where $\pi : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ is a permutation of maximal order, e.g. the cyclic permutation $\pi(i) = (i + 1) \bmod n$. Using this definition we can define a swap channel via:

$$S : \bigotimes_{i=1}^n \mathcal{B}(\mathbb{C}_i^2) \rightarrow \bigotimes_{i=1}^n \mathcal{B}(\mathbb{C}_i^2), \quad S(X_1 \otimes \dots \otimes X_n) = U_\pi^* (X_1 \otimes \dots \otimes X_n) U_\pi = X_{\pi(1)} \otimes \dots \otimes X_{\pi(n)}.$$

If we compose the channels T and S we get a root of the bistochastic CDC on the tensor system. To determine the order of that root, we analyze the behavior of that construction. Since the qubit root is of maximal order, it needs three steps of concatenation to completely depolarize a single qubit. The swap changes with the periodicity of the chain length n . Hence $3 \times n$ steps are needed to completely depolarize the chain. If we express this in terms of the dimension of the system $d = 2^n$, we find that the order the root is $k = 3 \log d$, which is a rather weak result, as we know that we can construct higher orders by embedding the classical roots.

4.4.2 Sum Spaces

Another possibility of embedding a qubit root into higher dimensions is to consider a sum of n qubit spaces $\mathcal{H} = \bigoplus_{i=1}^n \mathbb{C}_i^2 = \mathbb{C}_1^2 \oplus \dots \oplus \mathbb{C}_n^2$ where the dimension of the space grows linearly ($d = 2n$). We notice that the underlying operator algebra is given by $\mathcal{B}(\bigoplus_{i=1}^n \mathbb{C}_i^2) = \bigoplus_{i=1}^n \mathcal{B}(\mathbb{C}_i^2) \oplus \bigoplus_{i \neq j}^n \mathfrak{L}(\mathbb{C}_i^2, \mathbb{C}_j^2)$ ¹. We express an element of that algebra in the matrix notation as follows:

$$X = \left(\begin{array}{c|c|c} \mathbf{X}_{11} & \dots & \mathbf{X}_{1n} \\ \hline \vdots & \ddots & \vdots \\ \hline \mathbf{X}_{n1} & \dots & \mathbf{X}_{nn} \end{array} \right), \quad (4.22)$$

where $X_{ii} \in \mathcal{B}(\mathbb{C}_i^2)$ and $X_{i \neq j} \in \mathfrak{L}(\mathbb{C}_i^2, \mathbb{C}_j^2)$.

At first, we try to adopt the embedding of a root from the product spaces, which means that we compose the channel of a qubit root P and swap channel S as follows:

$$T(X_{11} \oplus \dots \oplus X_{nn}) = P(X_{11}) \oplus id_2(X_{22}) \oplus \dots \oplus id_2(X_{nn}). \quad (4.23)$$

Moreover, we find

$$S(X_{11} \oplus \dots \oplus X_{nn}) = \tilde{U}_\pi(X_{11} \oplus \dots \oplus X_{nn}) \tilde{U}_\pi^* = X_{\pi(11)} \oplus \dots \oplus X_{\pi(nn)}, \quad (4.24)$$

¹ $\mathfrak{L}$ denotes the set linear and bounded operators.

with $\pi : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ a permutation of maximal order, e.g. the cyclic permutation $\pi(i) = (i + 1) \bmod n$. By considering the definition of the channels T and S we find that its action is defined only on the subspace $\bigoplus_{i=1}^n \mathcal{B}(\mathbb{C}_i^2)$. Therefore, an additional channel M which projects an element of $\mathcal{B}(\bigoplus_{i=1}^n \mathbb{C}_i^2)$ to $\bigoplus_{i=1}^n \mathcal{B}(\mathbb{C}_i^2)$ is required. Its Kraus decomposition is:

$$M(X) = \sum_{i=1}^n \mathbb{1}_{ii} X \mathbb{1}_{ii}, \quad (4.25)$$

where $\mathbb{1}_{ii} \in \mathcal{B}(\mathbb{C}_i^2)^1$. The action of the composition of all three channels $K = STM$ in the case of three qubits is depicted in figure 4.3. We find that this channel depolarizes every subsystem after $k = \frac{3}{2}d$ steps. But, if we consider the eigenvalues of this composed channel,

$$K(X) = \begin{pmatrix} \mathbf{X}_{22} & 0 & 0 \\ 0 & \mathbf{X}_{33} & 0 \\ 0 & 0 & \mathbf{X}_{11}^* \end{pmatrix} \quad K^9(X) = \begin{pmatrix} \mathbf{X}_{11}^{***} & 0 & 0 \\ 0 & \mathbf{X}_{22}^{***} & 0 \\ 0 & 0 & \mathbf{X}_{33}^{***} \end{pmatrix}$$

Figure 4.3: The evolution of a three qubit system in case of the concatenation of the channel K . It is composed by a projective measurement which first reduces the input operator to an element of $\bigoplus_{i=1}^3 \mathcal{B}(\mathbb{C}_i^2)$. Then the qubit root (represented by the hammer) depolarizes the first subsystem, whereas the number of stars represents the degree of depolarization. Afterwards the swap channel interchanges the positions of the sum spaces. We find that after nine concatenation steps every subsystem is completely depolarized.

we find that it is *not* a root of the bistochastic CDC. Recall that the eigenvalue one is not degenerate for a bistochastic CDC. The channel K , however, preserves the completely mixed state of every subsystem $\mathcal{B}(\mathbb{C}_i^2)$. Therefore, the degeneracy of the eigenvalue one is equal to the number n of considered qubit systems.

To avoid the isolation between the systems we consider the following Kraus decomposition of a channel:

$$C(X) = \mathbb{1}_{21} X \mathbb{1}_{12} + \mathbb{1}_{11} X \mathbb{1}_{11} + \bigoplus_{i=3}^n (\mathbb{1}_{ii}) X (\mathbb{1}_{ii}). \quad (4.26)$$

This channel *copies* the subsystem $\mathcal{B}(\mathbb{C}_1^2)$ to the subsystem $\mathcal{B}(\mathbb{C}_2^2)$ while leaving the other systems untouched. If we compose this channel C with the projection M , the partial depolarizer T and a swap channel $\tilde{S}$,

$$\tilde{S}(X_{11} \oplus \dots \oplus X_{nn}) = U_\pi(X_{11} \oplus \dots \oplus X_{nn})U_\pi^* = X_{\pi(11)} \oplus X_{\pi(22)} \dots \oplus X_{\pi(nn)}, \quad (4.27)$$

which interchanges every but the first system, we find that this composition is a root. The process of depolarizing this channel $\tilde{K} = T\tilde{S}CM$ is depicted in figure 4.4. The order k of the root $\tilde{K}$ is equal to $n + 2$, since there are three steps of concatenation until the first subsystem is completely depolarized and then $n - 1$ steps until the first subsystem is copied to the rest. Since the preserved state of the channel is the completely mixed state

¹ The index of the $\mathbb{1}$ denotes here the position instead of its dimension.

$$\tilde{K}(X) = \left(\begin{array}{c|c|c|c} \begin{array}{c} \nearrow \\ \mathbf{X}_{11}^* \end{array} & 0 & 0 & 0 \\ \hline 0 & \mathbf{X}_{33} & 0 & 0 \\ \hline 0 & 0 & \mathbf{X}_{44} & 0 \\ \hline 0 & 0 & 0 & \mathbf{X}_{11} \end{array} \right) \quad \tilde{K}^2(X) = \left(\begin{array}{c|c|c|c} \begin{array}{c} \nearrow \\ \mathbf{X}_{11}^{**} \end{array} & 0 & 0 & 0 \\ \hline 0 & \mathbf{X}_{44} & 0 & 0 \\ \hline 0 & 0 & \mathbf{X}_{11} & 0 \\ \hline 0 & 0 & 0 & \mathbf{X}_{11}^* \end{array} \right)$$

Figure 4.4: The behavior of a four qubit system for the concatenation of the channel $\tilde{K} = T\tilde{S}CM$. First the projective measurement erases non diagonal entries. Then the channel C "copies" the entries X_{11} to X_{22} (illustrated by the dashed arrow). Afterwards the swap channel S causes a cyclic permutation and the channel T depolarizes the first subsystem.

on the first subsystem the resulting depolarizing channel is not bistochastic. If we consider the order k in dependency of the dimension d , we find $k = \frac{d}{2} + 2$, which means that the order is larger than the order of the constructed root in the product space, but still less than the order of an embedded maximal classical root.

CHAPTER 5

Clifford Channels as Roots

If we consider systems with larger input and output dimensions than in the qubit case, we find that the number of parameters for the state space and channels on it increases exponentially. Thus we restrict our approaches of constructing roots of high order to *Clifford channels*, which are characterized by their covariance with respect to Weyl systems [HW07, Hol02, Sch05]. Their main virtue is their simple parametrization even in higher dimensions, where we restrict the considered systems throughout this chapter to prime dimension. We discuss some approaches to construct Clifford roots and proof that $\frac{d^2-1}{2}$ is the tight upper bound for the order.

5.1 Weyl Operators

The Weyl operators are often referred to as generalized Pauli operators. Since the Pauli operators have so many specific properties, e.g. generating the $SU(2)$, being an unitary operator basis, fulfilling the commutation properties for angular momentum operators, etc., we need to be more specific about the property that is going to be generalized. An interesting way to consider the Pauli matrices is to map an element of the group $\mathbb{Z}_2 \times \mathbb{Z}_2$ in the following way [SVW]:

$$\begin{aligned}\mathbf{w}(0,0) &:= \mathbb{1} \\ \mathbf{w}(0,1) &:= \sigma_1 \\ \mathbf{w}(1,0) &:= \sigma_3 \\ \mathbf{w}(1,1) &:= i\sigma_2\end{aligned}\tag{5.1}$$

If we take a closer look at the map $\mathbf{w} : \mathbb{Z}_2 \times \mathbb{Z}_2 \mapsto M_2(\mathbb{C})$ we find the following relations:

$$\mathbf{w}(\eta + \xi) = f(\eta, \xi) \mathbf{w}(\eta) \mathbf{w}(\xi) \quad \text{with} \quad |f(\eta, \xi)| = 1,\tag{5.2}$$

where $\eta, \xi \in \mathbb{Z}_2^2$. This expresses the fact that $\eta \mapsto \mathbf{w}(\eta)$ is a *projective representation* of the group $\mathbb{Z}_2 \times \mathbb{Z}_2$, which we want to generalize to higher dimensions.

Considering how the map $\mathbf{w}$ can be generalized to $\mathbb{Z}_d \times \mathbb{Z}_d$, we restrict d to the cases where $\mathbb{Z}_d \times \mathbb{Z}_d$ forms the vector space $\mathbb{F}_d^2$, namely where d is prime. We refer to this space as *phase space*. For the definition of $\mathbf{w} : \mathbb{F}_d^2 \mapsto M_d(\mathbb{C})$ we use the notation $\eta^+ \in \mathbb{Z}_d$

and $\eta^- \in \mathbb{Z}_d$ for the components of an element $\eta \in \mathbb{F}_d^2$ and furthermore $\{|x\rangle | x = 1, \dots, d\}$ denotes a basis of $\mathbb{C}^d$:

$$\mathbf{w}(\eta^+, \eta^-) = U^{\eta^+} V^{\eta^-}, \quad (5.3)$$

where

$$V|x\rangle = \exp\left(\frac{2\pi i x}{d}\right)|x\rangle \quad \text{and} \quad U|x\rangle = |(x+1) \bmod d\rangle. \quad (5.4)$$

The matrices in (5.3) are the *Weyl operators*. We give an overview about some algebraic properties we frequently use in the following:

$$\mathbf{w}(\eta)\mathbf{w}(\xi) = e^{-i\beta(\eta,\xi)} \mathbf{w}(\eta + \xi) \quad (5.5)$$

$$\mathbf{w}(\eta)\mathbf{w}(\xi) = e^{i\sigma(\eta,\xi)} \mathbf{w}(\xi)\mathbf{w}(\eta) \quad (5.6)$$

with

$$\beta : \mathbb{F}_d^2 \times \mathbb{F}_d^2 \mapsto \mathbb{R}, \quad \beta(\eta, \xi) := \frac{2\pi\eta^+\xi^-}{d}, \text{ bilinear} \quad (5.7)$$

and

$$\sigma : \mathbb{F}_d^2 \times \mathbb{F}_d^2 \mapsto \mathbb{R}, \quad \sigma(\xi, \eta) := \beta(\eta, \xi) - \beta(\xi, \eta), \quad (5.8)$$

where $\sigma(\eta, \xi) = -\sigma(\xi, \eta)$, i.e. σ is a *symplectic* form.

By construction the Weyl operators are unitary. Thus we can express the $*$ -operation via:

$$\mathbf{w}^*(\eta)\mathbf{w}(\eta) = \mathbb{1} \quad \Rightarrow \quad \mathbf{w}(\eta)^* = e^{-\beta(\eta,\eta)} \mathbf{w}(-\eta). \quad (5.9)$$

Note that $\mathbf{w}(0,0)$ is in every dimension equal to the identity $\mathbb{1}_d$. The whole set of $\mathbf{w}(\eta_i)$ forms in M_d a unitary basis of operators which is orthogonal with respect to the Hilbert-Schmidt scalar product, i.e. $\text{tr}[\mathbf{w}(\eta)^*\mathbf{w}(\xi)] = d\delta_{\eta,\xi}$. Since $\mathbf{w}(0,0) = \mathbb{1}_d$, we find in particular $\text{tr}[\mathbf{w}(\eta)] = 0$ for all $\eta \neq (0,0)$. We use these properties in the next section to define a special class of channels.

5.2 Clifford Channels

In section 4.2.1 we have discussed the class of Pauli diagonal channels, i.e. channels which map a Pauli matrix to itself multiplied by a constant. We now consider a class of channels which is, compared to the Pauli diagonal channels, generalized in two ways. First, we define the channels on Weyl operators instead of Pauli matrices and secondly, we allow the channel to map a Weyl operator to another Weyl operator than itself. The following definition is a restricted version to prime dimensions and can be found in a more general context in [Güt08].

Definition 5.1 (Clifford Channel). Let $h : \mathbb{F}_d^2 \mapsto \mathbb{F}_d^2$ be a homomorphism, where d is prime. We call a channel $T : M_d(\mathbb{C}) \mapsto M_d(\mathbb{C})$ a **Clifford channel** if it is of the form:

$$T[\mathbf{w}(\eta)] = \lambda(\eta)\mathbf{w}(h(\eta)) \quad \lambda : \mathbb{F}_d^2 \mapsto \mathbb{C}. \quad (5.10)$$

Thus a Clifford channel maps Weyl operators to Weyl operators up to a constant and is completely determined by a homomorphism h , which can be represented by a 2×2 matrix over $\mathbb{F}_d$, and a complex valued function λ .

Of course, the property of T being unital and completely positive imposes some restriction on the choice of λ and h . Using the notation of Weyl operators, the unitality is expressed by:

$$T[\mathbf{w}(0)] = \lambda(0)\mathbf{w}(h(0)) = \mathbf{w}(0), \quad (5.11)$$

which induces the conditions $\lambda(0) = 1$ and $h(0) = 0$. The latter is fulfilled by every homomorphism due to its definition. To find the restriction caused by the complete positivity we refer to lemma 2.2, which states that $\forall a, \forall b : \sum_{ij} a_i^* T[b_i^* b_j] a_j \geq 0$ is a necessary and sufficient condition for T being positive. The following lemma and its proof can be found in [Güt08]:

Lemma 5.1. *A Clifford channel T is completely positive if and only if the matrix M with entries*

$$M_{i,j} = \lambda(\eta_i - \eta_j) \frac{e^{i\beta(\eta_i - \eta_j, \eta_j)}}{e^{i\beta(h(\eta_i) - h(\eta_j), h(\eta_j))}} \quad (5.12)$$

is positive.

Proof. We first to express the term $T[b_i^* b_j]$ in the notation of Weyl operators where we choose $b_i^* = \mathbf{w}(\eta_i)$ and the use the properties 5.5 and 5.9 of Weyl operators:

$$\begin{aligned} T[b_i^* b_j] &= T[\mathbf{w}(\eta_i) \mathbf{w}(\eta_j)^*] = e^{-i\beta(\eta_j, \eta_j)} T[\mathbf{w}(\eta_i) \mathbf{w}(-\eta_j)] \\ &= e^{-i\beta(\eta_j, \eta_j)} e^{-i\beta(\eta_i, -\eta_j)} T[\mathbf{w}(\eta_i - \eta_j)] \\ &= e^{i\beta(\eta_i - \eta_j, \eta_j)} \lambda(\eta_i - \eta_j) \mathbf{w}(h(\eta_i) + h(-\eta_j)) \\ &= e^{i\beta(\eta_i - \eta_j, \eta_j)} \lambda(\eta_i - \eta_j) e^{i\beta(h(\eta_i), h(-\eta_j))} \mathbf{w}(h(\eta_i)) \mathbf{w}(-h(\eta_j)) \\ &= e^{i\beta(\eta_i - \eta_j, \eta_j)} \lambda(\eta_i - \eta_j) e^{i\beta(h(\eta_i), h(-\eta_j))} e^{i\beta(h(\eta_j), h(\eta_j))} \mathbf{w}(h(\eta_i)) \mathbf{w}(h(\eta_j))^* \\ &= \underbrace{\lambda(\eta_i - \eta_j) \frac{e^{i\beta(\eta_i - \eta_j, \eta_j)}}{e^{i\beta(h(\eta_i) - h(\eta_j), h(\eta_j))}}}_{M_{ij}} \mathbf{w}(h(\eta_i)) \mathbf{w}(h(\eta_j))^*. \end{aligned} \quad (5.13)$$

If we now smartly choose $a_i = \tilde{a}_i \mathbf{w}(h(\eta_i))$ and put this in lemma 2.2 we find:

$$\begin{aligned} 0 \leq \sum_{ij} a_i^* T[b_i^* b_j] a_j &= \sum_{ij} \tilde{a}_i^* M_{ij} \underbrace{\mathbf{w}(h(\eta_i))^* \mathbf{w}(h(\eta_i)) \mathbf{w}(h(\eta_j))^* \mathbf{w}(h(\eta_j))}_{\mathbb{1}} \tilde{a}_j \\ &= \sum_{ij} \tilde{a}_i^* M_{ij} \tilde{a}_j \mathbb{1} \Rightarrow M \geq 0. \end{aligned} \quad (5.14)$$

Conversely, let $M \geq 0$. Then its coefficients $M_{i,j}$ can be written as $M_{i,j} = c_i^* c_j$. Since the Weyl operators form a basis, we can express arbitrary $a \in \mathcal{B}(\mathcal{H})$ and $b \in \mathcal{B}(\mathcal{H})$ via

$a = \sum_i \tilde{a}_i \mathbf{w}(\eta_i)$ and $b = \sum_i \tilde{b}_i \mathbf{w}(\eta_i)$. It follows:

$$\begin{aligned}
a^* T[b^* b] a &= \sum_{i,j,k,l} \tilde{a}_i^* \tilde{b}_j^* \tilde{b}_k \tilde{a}_l \mathbf{w}(\eta_i)^* T[\mathbf{w}(\eta_j)^* \mathbf{w}(\eta_k)] \mathbf{w}(\eta_l) \\
&= \sum_{i,j,k,l} \tilde{a}_i^* \tilde{b}_j^* \tilde{b}_k \tilde{a}_l \mathbf{w}(\eta_i)^* M_{j,k} \mathbf{w}(h(\eta_j)) \mathbf{w}(h(\eta_k))^* \mathbf{w}(\eta_l) \\
&= \sum_{i,j,k,l} \left(\tilde{a}_i \tilde{b}_j \tilde{c}_j \mathbf{w}(h(\eta_j))^* \mathbf{w}(\eta_i) \right)^* \underbrace{\tilde{b}_k \tilde{a}_l \tilde{c}_k \mathbf{w}(h(\eta_k))^* \mathbf{w}(\eta_l)}_{=: X_{k,l}} \\
&= \sum_{i,j,k,l} X_{j,i}^* X_{k,l} \geq 0.
\end{aligned} \tag{5.15}$$

□

Furthermore the following lemma characterizes the bistochasticity of a Clifford channel:

Lemma 5.2. *A Clifford channel T with $\text{rank } T \geq 2$ is bistochastic if the homomorphism h is invertible, i.e. $h \in GL(2, \mathbb{F}_d)$.*

Proof. Recall (2.36) for the definition of bistochasticity in the Heisenberg picture. Since the Weyl operators form a trace-free basis it follows:

$$\text{tr } X = \text{tr} \left(\sum_j a_j \mathbf{w}(\eta_j) \right) = a_0 d. \tag{5.16}$$

The trace of the image of T is given via:

$$\begin{aligned}
\text{tr}[T(X)] &= \text{tr} \left[T \left[\sum_j a_j \mathbf{w}(\eta_j) \right] \right] = \text{tr} \left[\sum_j a_j \lambda(\eta_j) \mathbf{w}(h(\eta_j)) \right] \\
&= a_0 d + \text{tr} \left[\sum_{j \neq 0} a_j \lambda(\eta_j) \mathbf{w}(h(\eta_j)) \right].
\end{aligned} \tag{5.17}$$

If we exclude the case where $\lambda(\eta_j) = 0$, $\forall \eta_j \neq 0$ we find that (5.16) and (5.17) are equal if $h(\eta_j) \neq 0 \forall \eta_j \neq 0$. □

Since the given example of the maximal qubit root in chapter 4.3 maps Pauli matrices to Pauli matrices, it represents a Clifford channel. We close this section writing the qubit root exemplarily in Clifford notation.

Example 5.1 (Maximal Clifford Qubit-Root). If we transform the representation matrix $\widetilde{T}^*$ from (4.19) from the Pauli basis into the Weyl basis we find:

$$\widetilde{T}^*_{\text{Weyl}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{-i}{2} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{i}{2} & 0 \end{pmatrix}. \tag{5.18}$$

This can be expressed in Clifford notation, i.e. the homomorphism $h : \mathbb{F}_2^2 \mapsto \mathbb{F}_2^2$ and the function $\lambda : \mathbb{F}_2^2 \mapsto \mathbb{C}$ in (5.10), as:

$$h = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \lambda(0,0) = 1, \quad \lambda(0,1) = 0, \quad \lambda(1,0) = \frac{i}{2}, \quad \lambda(1,1) = \frac{-i}{2}. \quad (5.19)$$

5.2.1 Weyl Diagonal Channels

To construct Weyl diagonal channels, we generalize the form of the Pauli diagonal channels by replacing the Pauli operators with Weyl operators:

$$T(X) = \sum_{\eta} \mu(\eta) \mathbf{w}(\eta)^* X \mathbf{w}(\eta) \quad \text{with} \quad \mu(\eta) \in \mathbb{R}. \quad (5.20)$$

Using (5.6), we see that these channels are indeed Weyl diagonal:

$$\begin{aligned} T[\mathbf{w}(\xi)] &= \sum_{\eta} \mu(\eta) \mathbf{w}(\eta)^* \mathbf{w}(\xi) \mathbf{w}(\eta) = \sum_{\eta} \mu(\eta) e^{i\sigma(\eta,\xi)} \mathbf{w}(\eta)^* \mathbf{w}(\eta) \mathbf{w}(\xi) \\ &= \lambda(\xi) \mathbf{w}(\xi), \end{aligned}$$

where

$$\lambda(\xi) = \sum_{\eta} e^{i\sigma(\eta,\xi)} \mu(\eta) = (\mathcal{F}\mu)(\xi). \quad (5.21)$$

is the discrete Fourier transform with respect to $\sigma(\eta,\xi)$ and coincides with the complex valued function λ in (5.10). The homomorphism $h : \mathbb{F}_d^2 \mapsto \mathbb{F}_d^2$ in (5.10) is for Weyl diagonal channels determined by $h = id_2$. By choosing the $\mu(\eta)$ as real numbers we find:

$$\lambda(\xi) = \sum_{\eta} e^{i\sigma(\eta,\xi)} \mu(\eta) = \sum_{\eta} e^{-i\sigma(\eta,-\xi)} \mu(\eta) = \overline{\lambda(-\xi)}. \quad (5.22)$$

Since there is no vector $\xi \in \mathbb{F}_d^2$ with $\xi = -\xi$ in prime dimensions $d \geq 3$ we see that if we choose $\lambda(\xi)$ to zero, $\lambda(-\xi)$ will be also zero. So far, the least possible non-zero dimension of the kernel, for a Weyl diagonal channel, is two.

Let us further investigate, if the complete positivity of T imposes more restrictions to the function λ that may influence the dimension of the kernel. Before we proceed to investigate these constraints, we introduce a theorem for the estimation of the eigenvalues of a complex square matrix.

Let $A \in M_d(\mathbb{C})$ and $R_i = \sum_{j \neq i} |a_{i,j}|$. We refer to the closed disc $B(a_{i,i}, R_i)$ in the complex plane as *Gershgorin disc*.

Theorem 5.1 (Gershgorin's Circle Theorem [Ger31]). *Every eigenvalue λ_j of A lies within at least one of the Gershgorin discs $B(a_{ii}, R_i)$.*

If we apply this theorem to hermitian matrices we find:

Corollary 5.1. *Any hermitian matrix A which is diagonally dominant, that is $|a_{i,i}| \geq \sum_{j \neq i} |a_{i,j}| = R_i$, and positive diagonal entries, i.e. $a_{i,i} > 0$, is positive.*

Proof. Since the eigenvalues of hermitian matrices are real we find by Gerschgorin's theorem that every eigenvalue λ_i lies inside the real intervall $[a_{i,i} - R_i, a_{i,i} + R_i]$ which is a closed subset of $\mathbb{R}^+$. $\square$

After introducing these tools, we consider the matrix M from lemma 2.2 to determine possible additional restrictions to the dimension of the kernel, imposed by the complete positivity of T . In the case of Weyl diagonal channels ($h = id_2$) the entries of M are reduced to:

$$M_{i,j} = \lambda(\eta_i - \eta_j) \Rightarrow M_{i,i} = \lambda(0) = 1. \quad (5.23)$$

If we consider the restrictions to λ from (5.22) we get that M is hermitian, i.e. $M_{i,j} = \overline{M_{j,i}}$, which is a necessary condition for M to be positive. If we choose the values for λ such that it fulfills the condition of lemma 5.1 for M being diagonally dominant, i.e. $\sum_{\eta \neq 0} |\lambda(\eta)| \leq 1$, we find that M is positive and hence T completely positive. Thus the choice of the dimension of the kernel of T is only restricted by (5.22) and we can find a suitable configuration of the $\mu(\eta)$ which represents a channel with a two dimensional kernel.

5.2.2 Reversible Clifford Channels

A channel is called *reversible*, if and only if it is represented by a homomorphism of the input and output algebra, i.e. a channel $T : \mathcal{B}(\mathcal{H}_{out}) \mapsto \mathcal{B}(\mathcal{H}_{in})$ with

$$T(XY) = T(X)T(Y) \quad \forall X, Y \in \mathcal{B}(\mathcal{H}_{out}). \quad (5.24)$$

We give an overview about the restrictions on the homomorphism h and the complex valued function λ in a Clifford channel $T(\mathbf{w}(\eta)) = \lambda(\eta)\mathbf{w}(h(\eta))$, while the corresponding derivations can be found in [Güt08]. A Clifford channel T is reversible if the following conditions are fulfilled:

- The homomorphism h is invertible, i.e. an isomorphism. Furthermore h needs to preserve the symplectic form σ , i.e. $\sigma(h(\eta), h(\xi)) = \sigma(\eta, \xi)$.
- The function λ is a phase, i.e. $|\lambda(\eta)| = 1 \quad \forall \eta \in \mathbb{F}_d^2$.
- Furthermore λ fulfills the relation $\lambda(\xi + \eta) = \lambda(\eta)\lambda(\xi) e^{i(\beta(\xi, \eta) - \beta(h(\xi), h(\eta)))}$. If we put this into (5.12) we find that $M_{i,j} = \frac{\lambda(\eta_i)}{\lambda(\eta_j)}$. Since this can be written as A^*A with $A_{1,k} = \{\lambda(\eta_1), \dots, \lambda(\eta_n)\}$ and $A_{l,k} = 0$ else, we find that M is positive and therefore this restriction ensures complete positivity of T .

If we express the property of symplecticity preserving in matrix notation for $h = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, we find

$$\begin{aligned} \sigma(h(\eta), h(\xi)) &= \sigma \left(\begin{pmatrix} a\eta^+ + b\eta^- \\ c\eta^+ + d\eta^- \end{pmatrix}, \begin{pmatrix} a\xi^+ + b\xi^- \\ c\xi^+ + d\xi^- \end{pmatrix} \right) \\ &= (ad - bc)\xi^- \eta^+ - (ad - bc)\xi^+ \eta^- \\ &= \sigma(\eta, \xi) \det h. \end{aligned} \quad (5.25)$$

We see that in matrix notation only those homomorphisms represent a reversible Clifford channel, whose determinant is equal to one, i.e. h is an element of the group $SL(2, \mathbb{F}_d)$.

5.3 Shifter and Destructor Approach

Considering section 4.3, we find that a maximal qubit root can be composed by a Pauli diagonal map and a suitable rotation. We generalize this concept to Clifford channels by constructing a root by a composition of two channels. First a Weyl diagonal map with least possible dimension of its kernel (the "destructor") and a reversible Clifford channel (the "shifter") which does not affect the algebraic multiplicity of the eigenvalue zero of the "destructor" but increases its geometric multiplicity. In other words the "destructor" part of the root should map in every step of concatenation some elements of the Weyl basis to zero while the "shifter" part shifts some of the remaining elements of that basis into the kernel of the "destructor". Figure 5.1 illustrates an example root for the qutrit ($\mathcal{H} = \mathbb{C}^3$) case, composed of the following two channels. First the "destructor" T_D :

$$T_D[\mathbf{w}(\eta)] = \sum_{\xi} \mu(\xi) \mathbf{w}(\xi)^* \mathbf{w}(\eta) \mathbf{w}(\xi) = \lambda_D(\eta) \mathbf{w}(h_D(\eta)), \quad (5.26)$$

where a suitable choice of the μ implies:

$$\lambda_D \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \lambda_D \begin{pmatrix} 0 \\ 2 \end{pmatrix} = \lambda_D \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \lambda_D \begin{pmatrix} 2 \\ 0 \end{pmatrix} = 0 \quad \text{and} \quad h_D = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (5.27)$$

Secondly a "shifter" channel T_S with:

$$T_S[\mathbf{w}(\eta)] = \lambda_D(\eta) \mathbf{w}(h_D(\eta)), \quad (5.28)$$

where we choose

$$h = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in SL(2, \mathbb{F}_3) \quad \text{and} \quad |\lambda(\eta)| = 1 \quad \forall \eta. \quad (5.29)$$

The behavior of that construction can be analyzed if we consider the concatenation of the channels T_D and T_S :

$$\begin{aligned} T_D [T_S[\mathbf{w}(\eta)]] &= T_D[\lambda_S(\eta) \mathbf{w}(h_S(\eta))] = \lambda_D(h_S(\eta)) \lambda_S(\eta) \mathbf{w}(h_D h_S(\eta)) \\ &\Rightarrow ||T_D [T_S[\mathbf{w}(\eta)]]|| = |\lambda_D(h_S(\eta))| ||\mathbf{w}(h_S(\eta))|| \\ &\Rightarrow ||(T_D T_S)^n[\mathbf{w}(\eta)]|| = \prod_{k=0}^{n-1} |\lambda_D(h_S^k(\eta))| ||\mathbf{w}(h_S^n(\eta))||. \end{aligned} \quad (5.30)$$

We see that the "shifting" part is determined by h_S while the elements that are mapped to zero are determined by λ_D . In figure 5.1 the order of the constructed root is three. Let us now investigate if this approach is suitable to construct roots of higher order than the classical embedded roots.

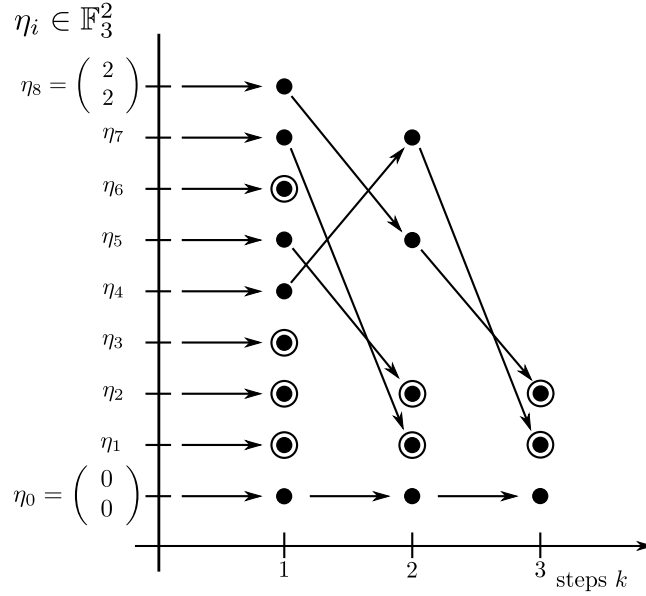


Figure 5.1: This qutrit root is constructed as a composition of a Weyl diagonal channel D as "destructor" and reversible Weyl channel S as "shifter". In every step of concatenation the "destructor" maps η_1, η_2, η_3 and η_6 to zero, which is illustrated by the circles. The channel S , however, maps in every concatenation step some elements of the remaining Weyl basis into the kernel, which is reflected by the arrows. The input system is completely depolarized when every vector but η_0 is mapped to zero. This is the case after three steps of concatenation.

5.3.1 An Upper Bound for the Approach

We start with the introduction of two useful definitions to characterize an upper bound:

Definition 5.2 (Group Order¹). Let G be a group. The **group order** of an element A of the group is defined as the smallest number l , such that $A^l = 1$.

Definition 5.3 (Orbit). Let V be a vector space over a finite field and $\phi : V \mapsto V$ an isomorphism. The **orbit** of $\xi \in V$ under ϕ is the set $O_\phi(\xi) := \{\phi^k(\xi) | k \in \mathbb{N}\}$. We refer to $|O_\phi(\xi)|$ as the *size* of the orbit.

Furthermore, we consider the bistochastic CDC in Clifford notation:

$$P[\mathbf{w}(\eta)] = \lambda_P(\eta) \mathbf{w}(h_P(\eta)) = \begin{cases} 1 & \eta = 0 \\ 0 & \eta \neq 0 \end{cases}. \quad (5.31)$$

Since the concatenation of the "shifter" and "destructor", i.e. $T_D T_S$, must coincide with (5.31) within a finite number of concatenation steps n and the homomorphism h_S is an element of $SL(2, \mathbb{F}_d)$, the term $\prod_{k=0}^{n-1} |\lambda_D(h_S^k(\eta))|$ in (5.30) needs to vanish for some n . This leads us to the the following lemma:

¹ This should not be confused with the order of a root

Lemma 5.3. *The bistochastic Clifford channel $T = T_D T_S$ is a root of the bistochastic CDC, if and only if there is at least one element of the kernel of T_D in every orbit O_{h_S} which is generated by h_S .*

Proof. With the previous used notations of the CDC and $T = T_D T_S$, it follows:

$$\prod_{k=0}^{n-1} |\lambda_D(h_S^k(\eta))| \|\mathbf{w}(h_S^n(\eta))\| = |\lambda_P(\eta)| \|\mathbf{w}(h_P(\eta))\| \quad (5.32)$$

$$\Leftrightarrow \exists n \in \mathbb{N} \forall \eta \neq 0 : \prod_{k=0}^{n-1} |\lambda_D(h_S^k(\eta))| = 0 \quad (5.33)$$

$$\Leftrightarrow \forall \eta \neq 0 \exists \alpha \in \{0, \dots, n-1\} : \lambda_D(h_S^\alpha(\eta)) = 0 \quad (5.34)$$

$$\Leftrightarrow \forall \eta \neq 0 \exists \eta_\alpha \in O_{h_S}(\eta) : \lambda_D(\eta_\alpha) = 0. \quad (5.35)$$

□

Using this lemma, we see that for an estimation of an upper bound of the root order it is sufficient to consider the largest possible orbit $O_{h_S, \max}$, where the estimation will be the largest, if we assume that λ_D maps only one element inside of $O_{h_S, \max}$ to zero. This situation is depicted in 5.2. We find that in this case the order coincides with the size if the orbit $O_{h_S, \max}$ to zero, i.e. $k = |O_{h_S, \max}|$. Following the previous reasoning we can

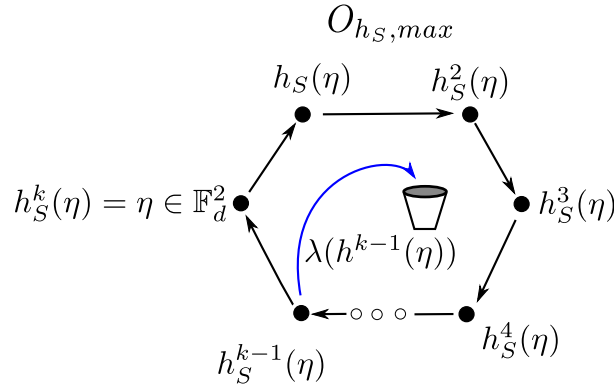


Figure 5.2: An estimation for the upper bound for the construction of a root with the "shift" and "destructor" approach can be obtained if we consider the largest orbit $O_{h_S, \max}$ of h_S of the "shifter" channel. We assume that only one element of the orbit lies inside the kernel of T_D . The number of steps k until every element of the orbit $O_{h_S, \max}$ is mapped to zero is $|O_{h_S, \max}|$.

obtain an upper bound for the root T , if we find an upper bound for the size of the largest orbit generated by h_S . The following lemma states a relation between the largest orbit $O_{h_S, \max}$ and the group order of h_S as an element of $SL(2, \mathbb{F}_d)$.

Lemma 5.4. *The group order l of an element h of $SL(2, \mathbb{F}_d)$ is larger or equal the size $|O_{h, \max}|$ of the largest orbit.*

Proof. Let l be the group order of $h \in SL(2, \mathbb{F}_d)$. Assume there is an orbit $\tilde{O}_h$ which size $\alpha := |\tilde{O}_h|$ is larger than the group order l , i.e. $\alpha > l$. It follows by the definition of the

size of an orbit, that α is the least possible number with:

$$\forall \eta_i \in \tilde{O}_h : h^\alpha(\eta_i) = \eta_i \quad \Rightarrow \quad h^\alpha = \mathbb{1}_2. \quad (5.36)$$

This is a contradiction to the definition of the group order l . $\square$

Using this lemma we see that the size of the largest orbit $|O_{h_S, \max}|$ is bounded by the order of h_S as an element of the finite group $SL(2, \mathbb{F}_d)$. The identification of the possible orders of the elements of $SL(2, \mathbb{F}_d)$ can be found in [App]. We find that the maximal order for an element of $SL(2, \mathbb{F}_d)$ is given by $2d$, with d the dimension of the underlying field. We conclude that $2d$ is an upper bound for the order of a root k constructed by the "shifter" and "destructor" approach.

5.4 General Clifford Roots

In the last section we found that the "shifter" and "destructor" approach, i.e. constructing a root as a composition of a Weyl diagonal and a reversible channel, is able to construct roots which order is asymptotically equivalent to the embedding of classical roots. Inside this section we will drop the restrictions of that approach and construct roots of order $\frac{d^2-1}{2}$, which turns out to be the highest possible order for Clifford channels in general.

Let us first consider the concatenation of an arbitrary bistochastic Clifford channel:

$$T[\mathbf{w}(\eta)] = \lambda(\eta) \mathbf{w}(h(\eta)) \quad (5.37)$$

$$\Rightarrow T^2[\mathbf{w}(\eta)] = T[\lambda(\eta) \mathbf{w}(h(\eta))] = \lambda(\eta) \lambda(h(\eta)) \mathbf{w}(h^2(\eta))$$

$$\Rightarrow T^n[\mathbf{w}(\eta)] = \prod_{k=0}^{n-1} \lambda(h^k(\eta)) \mathbf{w}(h^n(\eta)), \quad (5.38)$$

where $h \in GL(2, \mathbb{F}_d)$.¹ Finding the highest possible order of a Clifford root means to find the highest possible n , such that (5.38) is equivalent to (5.31). For this purpose we consider figure 5.3 which illustrates the evolution of a phase vector under the concatenation of T . The situation is similar to the concatenation of channels in the "shift" and "destructor" approach. In every step a vector of the phase space is mapped to another vector determined by h and then multiplied with the function λ . Here we can adopt the same argumentation as in lemma 5.3, namely that in every generated orbit $O_h(\eta_i)$ of h needs to lie a phase space vector η_α , which is mapped to zero by λ , i.e. $\lambda(\eta_\alpha) = 0$. This is equivalent to the vanishing of the product term for all η in (5.38), i.e. $\forall \eta \neq 0 : \prod_{k=0}^{n-1} \lambda(h^k(\eta)) = 0$. But we find two major differences to the concatenation of a "shifter" and a "destructor". First the choice of the complex function λ is not restricted as in the case of Weyl diagonal channels, i.e. $\lambda(\xi) = \overline{\lambda(-\xi)}$, and secondly the homomorphism h is an element of $GL(2, \mathbb{F}_d)$ instead of $SL(2, \mathbb{F}_d)$. We conclude that the highest root-order for bistochastic Clifford channels is determined by the least possible dimension of the kernel determined by λ and the size of the largest orbit for an element of $GL(2, \mathbb{F}_d)$.

¹ To determine an upper bound we exclude the case that h is not invertible, because the dimension of the kernel of the corresponding channel T is then at least $\frac{d^2-1}{2}$.

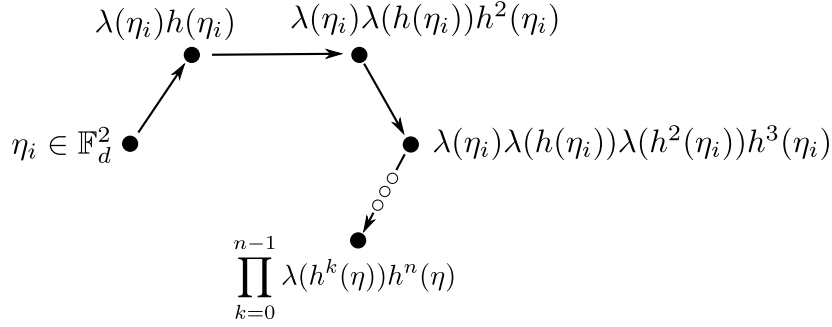


Figure 5.3: Evolution of a phase vector η_i under the concatenation of a bistochastic Clifford channel T .

5.4.1 Restrictions to the Kernel

Since the rank of a bistochastic Clifford channel T is completely determined by λ for invertible h , we recall the restrictions to λ we have derived so far. First $\lambda(0)$ needs to be equal to one, due to the unitality of T . The only additional restriction to the complex function λ for a general Clifford channel is induced by the positivity of the matrix M in (5.12). We know that a necessary condition for positivity is that M is hermitian. From this we derive further constraints on λ :

$$\begin{aligned}
 M_{i,j} &= M_{j,i}^* \\
 \lambda(\eta_i - \eta_j) \frac{e^{i\beta(\eta_i - \eta_j, \eta_j)}}{e^{i\beta(h(\eta_i) - h(\eta_j), h(\eta_j))}} &= \lambda^*(\eta_j - \eta_i) \frac{e^{i\beta(h(\eta_j) - h(\eta_i), h(\eta_i))}}{e^{i\beta(\eta_j - \eta_i, \eta_i)}} \\
 \frac{\lambda(\eta_i - \eta_j)}{\lambda^*(\eta_j - \eta_i)} &= \frac{e^{i\beta(h(\eta_j) - h(\eta_i), h(\eta_i))} e^{i\beta(h(\eta_i) - h(\eta_j), h(\eta_j))}}{e^{i\beta(\eta_j - \eta_i, \eta_i)} e^{i\beta(\eta_i - \eta_j, \eta_j)}} \\
 \frac{\lambda(\xi)}{\lambda^*(-\xi)} &= \underbrace{\frac{e^{i\beta(h(\xi), h(-\xi))}}{e^{i\beta(\xi, -\xi)}}}_{=: \alpha(\eta, \xi, -\xi)} \quad \text{with } \xi := \eta_i - \eta_j \\
 \lambda(\xi) &= \alpha(h, \xi, -\xi) \lambda^*(-\xi) \quad \text{with } |\alpha(h, \xi, -\xi)| = 1. \quad (5.39)
 \end{aligned}$$

We find the same restriction for the dimension of the kernel for arbitrary bistochastic Clifford channels as for Weyl diagonal channels, i.e. the least non-zero dimension of the kernel is two if $d \geq 3$. To assure the positivity of the eigenvalues of M we can repeat the argumentation of section 5.2.1 and choose the non-zero values of λ such that M is diagonally dominant, without any further restrictions to the kernel of T .

5.4.2 Orders of Elements of $GL(2, \mathbb{F}_d)$

In this section we probably use definitions from the realm of finite fields, which are unknown to the common reader. For a basic introduction to finite fields we recommend [LN03].

In the section 5.3.1 we have discussed the relation between the order of the homomorphism h and the size of the largest orbit $O_{h, \max}$ generated by h . We showed that the order of a possible root is bounded from above by the group order of h . Since for roots of high order, h is an element of $GL(2, \mathbb{F}_d)$, we investigate the maximal order of elements in that

group.

Theorem 5.2. *The largest group order of an element in $GL(2, \mathbb{F}_d)$ is $d^2 - 1$.*

Proof. To characterize the group order of an element $A \in GL(2, \mathbb{F}_d)$ it is important whether A can be diagonalized or not. Let us first assume that A is diagonalizable:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = P^{-1} \begin{pmatrix} e_1 & 0 \\ 0 & e_2 \end{pmatrix} P \quad (5.40)$$

$$A^k = (P^{-1})^k \begin{pmatrix} e_1^k & 0 \\ 0 & e_2^k \end{pmatrix} P^k. \quad (5.41)$$

Since changing the basis does not change the group order, it depends only on the eigenvalues of A or more precisely $\text{ord}(A) = \text{lcm}[\text{ord}(e_1), \text{ord}(e_2)]$ ¹. Hence we take a closer look at the eigenvalues, which are the roots of the characteristic polynomial $\det(A - \lambda)$. Here we distinguish two cases:

- i) The characteristic polynomial splits in $\mathbb{F}_d$. The maximal order of A is then $d - 1$, because every group order of an element of $\mathbb{F}_d$ divides $d - 1$, which is therefore the highest possible group order.
- ii) The characteristic polynomial is irreducible in $\mathbb{F}_d$ and splits in the *splitting field*, which is isomorphic to $\mathbb{F}_{d^2}$ [LN03]. The two eigenvalues are then elements of $\mathbb{F}_{d^2}$ and have the same group order. To find the maximal group order of A the characteristic polynomial needs to be a *primitive polynomial*, i.e. both zeros are *primitive elements* of the multiplicative group of $\mathbb{F}_{d^2}$. The group order of A is then $d^2 - 1$.

Now assume that A cannot be diagonalized. This implies that A has one degenerate eigenvalue. This implies that the characteristic polynomial splits in $\mathbb{F}_d$. Thus A is similar to the Jordan block:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = P^{-1} \begin{pmatrix} e_1 & 1 \\ 0 & e_1 \end{pmatrix} P. \quad (5.42)$$

If we take A to the power of k it follows:

$$A^k = (P^{-1})^k \begin{pmatrix} e_1^k & e_1^k k \\ 0 & e_1^k \end{pmatrix} P^k. \quad (5.43)$$

Since for an arbitrary element e_1 of $\mathbb{F}_d$ the largest number $\alpha \in \mathbb{F}_d$ such that $\alpha e_1 = 1$ is given by the dimension of the field, it follows that the maximal group order of A is given by $\text{ord}(A) = \text{lcm}[d, (d - 1)] = d(d - 1) = d^2 - d$.

Summarizing all three cases we see that $d^2 - 1$ is the highest possible group order and can be reached by choosing A such that $\det(A - \lambda)$ is a primitive polynomial over $\mathbb{F}_d$. $\square$

Although the maximal group order of an element $h \in GL(2, \mathbb{F}_d)$ coincides with the maximal order of a root of a quantum CDC, it is not clear whether a maximal h constructs

¹ lcm = least common multiple

a maximal root T , since also the number of generated orbits determines the order of a root. To characterize the orbits that are generated by a homomorphism h with maximal order $d^2 - 1$, we need the following two lemmata as a preparation.

Lemma 5.5. *Let α be the size of the orbit $O_\phi(\xi)$. Then $(\phi^{-1})^k(\xi) = \phi^{\alpha-k}(\xi)$.*

Proof. If α is the size of the orbit it follows by definition that

$$\phi^\alpha(\xi) = \xi. \quad (5.44)$$

The lemma follows by multiplying with the inverse. $\square$

Lemma 5.6. *If an element η lies inside an orbit of another element ξ both orbits coincide, i.e. $\eta \in O_\phi(\xi) \Rightarrow O_\phi(\xi) = O_\phi(\eta)$.*

Proof. Let $\tilde{\xi} \in O_\phi(\xi)$, i.e. $\exists \tilde{k} : \phi^{\tilde{k}}(\xi) = \tilde{\xi}$. Since η lies in the orbit of ξ there is a k such that:

$$\phi^{\tilde{k}+k}(\xi) = \eta \quad (5.45)$$

$$\tilde{\xi} = \phi^{\tilde{k}}(\xi) = (\phi^{-1})^k(\eta) \stackrel{\text{lemma 5.5}}{=} \phi^{\alpha-k}(\eta), \quad (5.46)$$

with α as in Lemma 5.5. Therefore $\tilde{\xi} \in O_\phi(\eta)$.

Conversely, let $\tilde{\eta} \in O_\phi(\eta)$, i.e. $\exists \tilde{k} : \phi^{\tilde{k}}(\eta) = \tilde{\eta}$. It follows, that there is a k such that:

$$\tilde{\eta} = \phi^{\tilde{k}}(\eta) = \phi^{\tilde{k}}(\phi^k(\xi)) = \phi^{\tilde{k}+k}(\xi), \quad (5.47)$$

and therefore $\tilde{\eta} \in O_\phi(\xi)$. $\square$

Theorem 5.3. *An element $A \in GL(2, \mathbb{F}_d)$ with order $d^2 - 1$ spans only one orbit apart from the orbit of the zero vector.*

We notice that a necessary condition for an element $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ of group order $d^2 - 1$ is that the characteristic polynomial is irreducible and therefore $c, b \neq 0$. Furthermore, the underlying vector space consists of d^2 elements. In order to show that there is only one orbit apart from the orbit of the zero vector, it suffices to find an orbit with $d^2 - 1$ distinct elements (see Lemma 5.6).

Proof. We consider the orbit of the vector $e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ under the map A . Assume that the size α of $O_A(e_1)$ is smaller than $d^2 - 1$, i.e.

$$\exists \tilde{\alpha} < d^2 - 1 : A^{\tilde{\alpha}}(e_1) = e_1. \quad (5.48)$$

Then it follows:

$$\forall k \in \mathbb{N} : \underbrace{A^{\tilde{\alpha}} A^k(e_1)}_{\in O_A(e_1)} = A^{\alpha}(e_1). \quad (5.49)$$

Since this holds for all k and with $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} a \\ c \end{pmatrix}$, there are two linear independent vectors inside $O_A(e_1)$ and it follows that $A^{\tilde{\alpha}} = \mathbb{1}$. This contradicts the condition that the order of A is $d^2 - 1$. $\square$

Summarizing this section we say that we can find a homomorphism $h \in GL(2, \mathbb{F}_d)$ in (5.38) that spans, apart from the trivial orbit of the zero vector, only one orbit of size $d^2 - 1$ inside the phase space.

5.4.3 Maximal Order of Clifford Roots

We now use the results of the sections 5.4.1 and 5.4.2 to characterize the upper bound for the construction of a root of the bistochastic CDC with bistochastic Clifford channels in prime dimensions. We recall that the concatenation of an arbitrary bistochastic qubit channel T is of the form:

$$T^n[\mathbf{w}(\eta)] = \prod_{k=0}^{n-1} \lambda(h^k(\eta)) \mathbf{w}(h^n(\eta)), \quad (5.50)$$

where $h \in GL(2, \mathbb{F}_d)$ and $\lambda : \mathbb{F}_d^2 \mapsto \mathbb{C}$.

Theorem 5.4. $\frac{d^2-1}{2}$ is the upper bound for a Clifford root of the bistochastic CDC.

Proof. Considering lemma 5.3 we find that a necessary and sufficient condition for bistochastic Clifford channel T being a root of a CDC is that in every orbit which is spanned by h in (5.50) lies at least one element of the kernel of T . Clearly the order k of the root will be maximal, if the dimension of the kernel of T is minimal. Due to section 5.4.1 the minimal dimension of the kernel is two, where both elements of the kernel are inverse elements of the phase space, i.e. $\lambda(\eta_j) = \lambda(-\eta_j) = 0$. Since the sizes $|O_h(\eta_j)|$ and $|O_h(-\eta_j)|$ of the orbits of both elements generated by h coincide, i.e.

$$h^\alpha(\eta_j) = \eta_j \quad \Leftrightarrow \quad h^\alpha(-\eta_j) = -\eta_j, \quad (5.51)$$

we find that T is a root if and only if the orbits of both elements cover the whole phase space, i.e. $|O_h(\eta_j)| = \frac{d^2-1}{2}$ or $|O_h(\eta_j)| = d^2 - 1$. Both cases need the same steps of concatenation to depolarize the complete system, i.e. $\frac{d^2-1}{2}$, which is therefore an upper bound for the order of a root. Considering section 5.4.2 we find that this bound can be reached by choosing the homomorphism h as an element of $GL(2, \mathbb{F}_d)$ with maximal group order (see theorem 5.3). This element generates only one orbit of size $d^2 - 1$ and represents together with a suitable λ a maximal Clifford root. $\square$

Finally we see that the maximal order for Clifford roots is asymptotically the same as the highest possible order for general quantum channels.

CHAPTER 6

Maximal Quantum Roots

In this chapter we present an approach to construct roots of the highest possible order k , i.e. $k = d^2 - 1$. This approach is based on the idea in section 3.3.2, where we constructed maximal roots for classical systems. Furthermore we introduce some applications of maximal roots in the context of finitely correlated states and quantum memory channels.

6.1 Roots via Perturbation

If we recall section 3.3.2, we find that the principle to construct a maximal root of the classical bistochastic CDC is to add an ε -weighted maximal Jordan block of the eigenvalue zero to the Jordan decomposition of the CDC. Adopting this to quantum channels, the key is to find a suitable ε in an appropriate Jordan basis, such that the resulting map is completely positive and therefore represents a channel. We baptize this construction as *roots via perturbation*.

6.1.1 Properties of the Jordan Basis

Let us first consider the properties of the Jordan basis of a maximal root. We use the same notation for the Jordan decomposition as in section 3.3.2, i.e. writing the channel

$$R = \begin{pmatrix} 1 & 0 \\ 0 & J(0) \end{pmatrix} \in M_{d^2}(\mathbb{C}) \quad J(0) := \begin{pmatrix} 0 & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{pmatrix} \in M_{d^2-1}(\mathbb{C}), \quad (6.1)$$

using elements of an operator basis $B = \{A_1, \dots, A_{d^2}\}$ and its dual $B^* = \{A^1, \dots, A^{d^2}\}$ w.r.t. the Hilbert-Schmidt scalar product, $\text{tr}(A_i^* A^j) = \delta_{i,j}$, as follows:

$$R(X) = \underbrace{A_1 \text{tr}(A^1 X)}_{=: P(X)} + \underbrace{\sum_{\alpha=2}^{d^2-1} A_\alpha \text{tr}(A^{\alpha+1} X)}_{=: J(X)}. \quad (6.2)$$

The n -th concatenation of R is given by:

$$R^n(X) = A_1 \operatorname{tr}(A^{1*} X) + \sum_{\alpha=2}^{d^2-n} A_\alpha \operatorname{tr}(A^{\alpha+n*} X). \quad (6.3)$$

In particular it follows that after $d^2 - 1$ concatenation steps only the first term is left, i.e.

$$R^{d^2-1}(X) = A_1 \operatorname{tr}(A^{1*} X). \quad (6.4)$$

Since R is a maximal root of the CDC (2.30), we find that $A_1 = \mathbb{1}_d$ and $A^1 = \frac{\mathbb{1}_d}{d}$. Therefore, every other element of the basis and its dual is trace-free, i.e. $\forall i \neq 1 : \operatorname{tr} A^i = \operatorname{tr} A_i = 0$. The Choi matrix of the channel R is given by:

$$\begin{aligned} \sigma_R &= \frac{1}{d} \sum_{i,j} R(|i\rangle\langle j|) \otimes |i\rangle\langle j| \\ &= \frac{1}{d} \sum_{i,j} \left(A_1 \operatorname{tr}(A^{1*} |i\rangle\langle j|) + \sum_{\alpha=2}^{d^2-1} A_\alpha \operatorname{tr}(A^{\alpha+1*} |i\rangle\langle j|) \right) \otimes |i\rangle\langle j| \\ &= \frac{1}{d} \sum_{i,j} \left(A_1 A_{j,i}^{1*} + \sum_{\alpha=2}^{d^2-1} A_\alpha A_{j,i}^{\alpha+1*} \right) \otimes |i\rangle\langle j| \\ &= \frac{1}{d} A_1 \otimes \overline{A^1} + \frac{1}{d} \sum_{\alpha=2}^{d^2-1} A_\alpha \otimes \overline{A^{\alpha+1}} = \frac{\mathbb{1}}{d^2} + \frac{1}{d} \sum_{\alpha=2}^{d^2-1} A_\alpha \otimes \overline{A^{\alpha+1}}. \end{aligned} \quad (6.5)$$

Recalling that the hermiticity of the Choi matrix σ_R is a necessary condition for R being a channel, we find the following necessary conditions for the elements of the bases B and B^* :

$$\begin{aligned} \sigma_R &= \sigma_R^* \\ \Leftrightarrow \quad \frac{\mathbb{1}}{d^2} + \frac{1}{d} \sum_{\alpha=2}^{d^2-1} A_\alpha \otimes \overline{A^{\alpha+1}} &= \frac{\mathbb{1}}{d^2} + \frac{1}{d} \sum_{\alpha=2}^{d^2-1} A_\alpha^* \otimes \overline{A^{\alpha+1}}^* \\ \Rightarrow \quad A_\alpha &= \lambda A_\alpha^* \text{ and } A^\alpha = \bar{\lambda} A^\alpha, \quad |\lambda| = 1, \end{aligned} \quad (6.6)$$

$$\Rightarrow \quad A_\alpha = \lambda A_\alpha^* \text{ and } A^\alpha = \bar{\lambda} A^\alpha, \quad |\lambda| = 1, \quad (6.7)$$

i.e. every $A_i \in B \setminus A_1$ and $A^i \in B^* \setminus A^1$ is hermitian up to a phase.

6.1.2 The Bound on ε

After we derived the properties of a suitable Jordan basis such that the channel is bistochastic and the corresponding Choi matrix is hermitian, we now consider the positivity of the Choi matrix. For this purpose we weight the Jordan block in (6.2) with ε , i.e.

$$R_\varepsilon(X) = A_1 \operatorname{tr}(A^{1*} X) + \varepsilon \sum_{\alpha=2}^{d^2-1} A_\alpha \operatorname{tr}(A^{\alpha+1*} X), \quad (6.8)$$

and consider the corresponding Choi matrix:

$$\sigma_{R_\varepsilon} = \underbrace{\frac{\mathbb{1}}{d^2}}_{=\sigma_P} + \varepsilon \underbrace{\frac{1}{d} \sum_{\alpha=2}^{d^2-1} A_\alpha \otimes \overline{A^{\alpha+1}}}_{=: \sigma_J}, \quad (6.9)$$

where σ_P is the Choi matrix of the bistochastic CDC and σ_J the Choi matrix of the ε -weighted Jordan block. Due to the fact that the A_i and A^i are hermitian up to a phase and trace-free, σ_J will be hermitian and trace-free as well. Therefore σ_J is unitarily diagonalizable and its eigenvalues sum to zero. The eigenvalues of σ_{R_ε} can be determined via:

$$\begin{aligned} \sigma_{R_\varepsilon} &= \frac{\mathbb{1}}{d^2} + \varepsilon \frac{1}{d} \sum_{\alpha=2}^{d^2-1} A_\alpha \otimes \overline{A^{\alpha+1}} \\ &= \frac{\mathbb{1}}{d^2} + \varepsilon U^* \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_{d^2} \end{pmatrix} U \end{aligned} \quad (6.10)$$

$$= U^* \begin{pmatrix} \frac{1}{d^2} + \varepsilon \lambda_1 & 0 & \dots & 0 \\ 0 & \frac{1}{d^2} + \varepsilon \lambda_2 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \dots & \frac{1}{d^2} + \varepsilon \lambda_{d^2} \end{pmatrix} U, \quad (6.11)$$

where $\sum_{i=0}^{d^2} \lambda_i = 0$. Therefore, the positivity of σ_{R_ε} is assured, if we choose an ε such that $\forall i : |\varepsilon| \lambda_i \geq -\frac{1}{d^2}$.

In what follows, we consider two different norms on the Choi matrices and in terms of these we derive upper bounds for $|\varepsilon|$ such that the Choi matrix σ_{R_ε} is positive and thus the channel R_ε completely positive. First, we consider the *spectral norm* $\|\cdot\|_2$ of the σ_J , i.e. $\|\sigma_J\|_2 = \sqrt{\lambda_{\max}(\sigma_J^* \sigma_J)}$, and find the following lemma:

Lemma 6.1. *Let $A_i \in B$ and $A^i \in B^*$ be a Jordan basis and its dual, fulfilling the previously derived constraints, i.e. $A_1 = \mathbb{1}_d$, $\text{tr } A_i = d\delta_{1,i}$ and $\forall i \geq 2 : A_i = \alpha A_i^*$ with $|\alpha| = 1$. Then the map $R(X) = P(X) + \varepsilon J(X) = \frac{\mathbb{1}_d}{d} \text{tr } X + \varepsilon \sum_{\alpha=2}^{d^2-1} A_\alpha \text{tr}(A^{\alpha+1*} X)$ is completely positive if $|\varepsilon| \leq \frac{1}{\|\sigma_J\|_2 d^2}$.*

Proof. Since the matrix σ_J is hermitian, it follows $\sqrt{\lambda_{\max}(\sigma_J^* \sigma_J)} = |\lambda_{\max}(\sigma_J)|$. Therefore, $\|\varepsilon \sigma_J\|_2 \leq \frac{1}{d^2}$ implies $|\varepsilon \lambda_{\max}(\sigma_J)| \leq \frac{1}{d^2}$. Considering (6.11) we see immediately that this implies the positivity of σ_{R_ε} . $\square$

Another possible way to characterize the positivity of σ_J arises, if we use the Hilbert-Schmidt norm, i.e. $\|\sigma_J\|_{HS} = \sqrt{\text{tr}(\sigma_J^* \sigma_J)}$, to express an upper bound on $|\varepsilon|$.

Corollary 6.1. *Let R be a map as in lemma 6.1. It is completely positive if $|\varepsilon| \leq \frac{\sqrt{2}}{\|\sigma_J\|_{HS} d^2}$.*

Proof. Considering lemma 6.1 it is sufficiently to show $\|\sigma_J\|_{HS}^2 = \sum_i \lambda_i^2 \leq \frac{2}{d^4}$ implies $|\lambda_{\max}(\sigma_J)| \leq \frac{1}{d^2}$. We prove this by contradiction:

Assume $\|\sigma_J\|_{HS}^2 \leq \frac{2}{d^4}$ and $|\lambda_{\max}(\sigma_J)| > 1/d^2$. Since the trace of σ_J is zero, i.e. $\text{tr } \sigma_J = \sum_i \lambda_i(\sigma_J) = 0$, we find that also $\sum_{i \neq \max} |\lambda_i(\sigma_J)| > 1/d^2$. In the case that the sum consists only of one element, i.e. $\sum_{i \neq \max} |\lambda_i(\sigma_J)| = |\lambda_j(\sigma_J)| > \frac{1}{d^2}$, it follows $\|\sigma_J\|_{HS}^2 = \sum_i \lambda_i^2 = \lambda_{\max}^2 + \lambda_j^2 > \frac{2}{d^4}$. $\square$

We use this corollary for an explicit calculation of $|\varepsilon|$ in terms of the dimension. We consider the case where the Jordan basis B in (6.8) is an orthogonal basis, i.e. $\text{tr}(A_i^* A_j) = \text{tr}(A^{i*} A^j) = \delta_{i,j}$:

$$\|\sigma_J\|_{HS}^2 = \text{tr}(\sigma_J^* \sigma_J) = \frac{1}{d^2} \text{tr} \left(\sum_{\alpha=2}^{d^2-1} A_\alpha^* \otimes \overline{A^{\alpha+1}}^* \sum_{\beta=2}^{d^2-1} A_\beta \otimes \overline{A^{\beta+1}} \right) \quad (6.12)$$

$$= \frac{1}{d^2} \text{tr} \left(\sum_{\alpha, \beta=2}^{d^2-1} A_\alpha^* A_\beta^* \otimes \overline{A^{\alpha+1}}^* \overline{A^{\beta+1}}^* \right) \quad (6.13)$$

$$= \frac{1}{d^2} \sum_{\alpha, \beta=2}^{d^2-1} \text{tr}(A_\alpha^* A_\beta^*) \text{tr}(\overline{A^{\alpha+1}}^* \overline{A^{\beta+1}}^*) = \frac{d^2 - 2}{d^2} = 1 - \frac{2}{d^2}. \quad (6.14)$$

Using corollary 6.1 we find the following bound on $|\varepsilon|$ for orthogonal bases:

$$\varepsilon^2 \leq \frac{2}{\|\sigma_J\|_{HS}^2 d^4} = \frac{2}{(1 - \frac{2}{d^2}) d^4} = \frac{2}{d^4 - 2d^2} \Rightarrow |\varepsilon| \leq \frac{\sqrt{2}}{d\sqrt{d^2 - 2}}. \quad (6.15)$$

The following theorem summarizes the results of this chapter:

Theorem 6.1. *Let $B := \{A_1 = \mathbb{1}_d, A_2, \dots, A_{d^2}\}$ be a basis of the hermitian $d \times d$ operators and $B^* = \{A^1 = \frac{\mathbb{1}_d}{d}, A^2, \dots, A^{d^2}\}$ its dual basis. Then there is an ε , such that the map $R : M_d \mapsto M_d$, $R(X) = \frac{\mathbb{1}_d}{d} \text{tr } X + \varepsilon \sum_{\alpha=2}^{d^2-1} A_\alpha \text{tr}(A^{\alpha+1*} X)$ is completely positive and therefore a root of the bistochastic CDC of order $d^2 - 1$ with Kraus rank $d^2 - 1$.*

Proof. The map R is unital, i.e. $R(\mathbb{1}_d) = \mathbb{1}_d$, due to the fact that $\text{tr } A^i = d\delta_{1,i}$. Considering concatenations of R (6.3) we find that R is completely depolarizing not before $d^2 - 1$ concatenation steps, i.e. $R^{d^2-1}(X) = \frac{\mathbb{1}_d}{d} \text{tr } X$. The corresponding Choi matrix σ_R is hermitian (6.7) and with $\varepsilon \leq \frac{1}{\|\sigma_J\|_{HS}^2 d^2}$, as in lemma 6.1, positive. The Kraus rank is determined by the eigenvalues of σ_{R_ε} in (6.11). Due to the continuity of the eigenvalues of σ_{R_ε} as functions of ε , i.e. $\lambda_{R_\varepsilon} = \frac{1}{d^2} + \varepsilon \lambda_i$, and the condition $\text{tr}(\varepsilon \lambda_i) = 0$ we find that there is an ε , where $\exists i : \varepsilon \lambda_i = -\frac{1}{d^2}$. For this ε the Kraus rank of the root is $d^2 - 1$. $\square$

An open problem is the question, if it is possible to construct maximal roots with a smaller Kraus rank, i.e. a degeneracy of the eigenvalue zero of the Choi matrix. This could be possible in dimension $d \leq 3$, since the investigations of the qubit case showed, that for maximal qubit roots one needs at least three Kraus operators (see section 4.2.1). We close the section by applying this approach to qubits:

Example 6.1. We choose the Pauli matrices as basis B :

$$R_\varepsilon(X) = \frac{\mathbb{1}_d}{d} \text{tr } X + \varepsilon \left(\tilde{\sigma}_1 \text{tr} \left(\tilde{\sigma}^{2*} X \right) + \tilde{\sigma}_2 \text{tr} \left(\tilde{\sigma}^{3*} X \right) \right). \quad (6.16)$$

Since the Pauli operators form an orthogonal basis, we use (6.15) to calculate the upper bound on ε :

$$\varepsilon = \frac{\sqrt{2}}{d\sqrt{d^2-2}} \stackrel{d=2}{=} \frac{1}{2}. \quad (6.17)$$

If we consider the matrix

$$R_{\frac{1}{2}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (6.18)$$

we find that it is equal to the maximal qubit root (4.19), we previously constructed in chapter 4.

6.2 Applications

Now that we derived a way to construct roots of maximal order, we consider some applications. First, we consider finitely correlated states, where a maximal root is used to construct a k -dependent state. Furthermore, we consider the class of memory channels. We find that a maximal root is a strictly forgetful memory channel without input.

6.2.1 Finitely Correlated States

The general concept of *finitely correlated states* is explained in [FNW92]. To understand the application of the maximal roots in this context, we first recall that every channel T admits a Stinespring representation (see theorem 2.3), i.e.

$$T : \mathcal{B}(\mathcal{H}_2) \mapsto \mathcal{B}(\mathcal{H}_1), \quad T(X) = V^*(X \otimes \mathbb{1}_k)V \quad \forall X \in \mathcal{B}(\mathcal{H}_2), \quad (6.19)$$

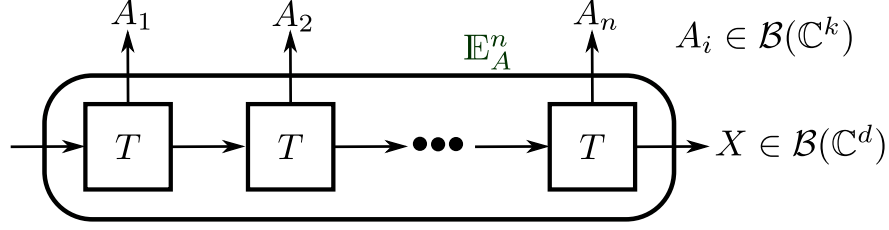
where $V : \mathbb{C}^d \mapsto \mathbb{C}^{d'} \otimes \mathbb{C}^k$ is an isometry, that is $V^*V = \mathbb{1}$. Since we want to concatenate the channel T , input dimension $d = \dim(\mathcal{H}_1)$ and output dimension $d' = \dim(\mathcal{H}_2)$ will be equal. Furthermore, the dimension k of the ancilla system is equal to the Kraus rank of T . Keeping this representation in mind, we define the following map:

$$\mathbb{E}_A : \mathcal{B}(\mathbb{C}^d) \otimes \mathcal{B}(\mathbb{C}^k) \mapsto \mathcal{B}(\mathbb{C}^d), \quad \mathbb{E}_A(X) = V^*(X \otimes A)V, \quad (6.20)$$

with V as in (6.19). Note that $\mathbb{E}_1 = T$. We introduce the following notation for the concatenation of the map:

$$\mathbb{E}_{A_1} \circ \mathbb{E}_{A_2} \circ \dots \circ \mathbb{E}_{A_n}(X) := T(A_1 \otimes T(A_2 \otimes T(\dots T(A_n \otimes X)\dots))). \quad (6.21)$$

Considering figure 6.1, we see that the n -th concatenation of $\mathbb{E}_A$ maps from $\mathcal{B}(\mathbb{C}^k)^{\otimes n} \otimes \mathcal{B}(\mathbb{C}^d)$ to $\mathcal{B}(\mathbb{C}^d)$. In what follows, we use the concatenation of the map $\mathbb{E}_A$ and a state ρ to define

Figure 6.1: Concatenation of the map $\mathbb{E}_A$

a functional $\omega_n : \mathcal{B}(\mathbb{C}^k)^{\otimes n} \mapsto \mathbb{C}$ via:

$$\omega_n(A_1 \otimes A_2 \otimes \dots \otimes A_n) = \text{tr}(\rho \mathbb{E}_{A_1} \circ \mathbb{E}_{A_2} \circ \dots \circ \mathbb{E}_{A_n}(\mathbb{1})) \quad \text{with } \rho \in \mathcal{S}(\mathbb{C}^d). \quad (6.22)$$

Due to the unitality of T , we can extend this functional to the positive half chain $\mathcal{A}_+ := \bigotimes_{i=0}^{\infty} \mathcal{B}(\mathbb{C}_i^k)$ via:

$$\omega_{n+1}(A_{[1,n]} \otimes \mathbb{1}) = \text{tr} \left(\rho \mathbb{E}_{A_1} \circ \mathbb{E}_{A_2} \circ \dots \circ \mathbb{E}_{A_n} \circ \underbrace{\mathbb{E}_{\mathbb{1}}(\mathbb{1})}_{\mathbb{1}} \right) \quad (6.23)$$

$$= \text{tr}(\rho \mathbb{E}_{A_1} \circ \mathbb{E}_{A_2} \circ \dots \circ \mathbb{E}_{A_n}(\mathbb{1})) = \omega_n(A_{[1,n]}). \quad (6.24)$$

Furthermore, if we choose ρ as an invariant state of T , i.e. $\text{tr}(\rho \mathbb{E}_{\mathbb{1}}(X)) = \text{tr}(\rho X)$, we can extend the functional also to the negative half-chain $\mathcal{A}_- := \bigotimes_{i=-\infty}^0 \mathcal{B}(\mathbb{C}_i^k)$ via:

$$\omega_{n+1}(\mathbb{1} \otimes A_{[1,n]}) = \text{tr} \left(\underbrace{\rho \mathbb{E}_{\mathbb{1}}}_{\rho} \circ \mathbb{E}_{A_1} \circ \mathbb{E}_{A_2} \circ \dots \circ \mathbb{E}_{A_n}(\mathbb{1}) \right) \quad (6.25)$$

$$= \omega_n(A_{[1,n]}). \quad (6.26)$$

Therefore the functional can be extended to the infinite spin chain $\mathcal{A} := \bigotimes_{i=-\infty}^{\infty} \mathcal{B}(\mathbb{C}_i^k)$, where we assume $\mathcal{A}$ to be a *quasi-local algebra*, i.e. an algebra on the infinite spin chain generated by algebras of finite subsets $\mathcal{A}_\Lambda := \bigotimes_{z \in \Lambda} \mathcal{A}_z$ on finite chain elements. If we now define the *shift operator* σ by setting

$$\sigma : \mathcal{A}_\Lambda \mapsto \mathcal{A}_{\Lambda+1} \quad \sigma(A_1 \otimes \dots \otimes A_n \otimes \mathbb{1}) = \mathbb{1} \otimes A_1 \otimes \dots \otimes A_n, \quad (6.27)$$

we find that ω is a *translation invariant state*, i.e. $\omega(\mathcal{A}) = \omega(\sigma \circ \mathcal{A})$. Moreover, due to its construction by T and ρ in (6.22) we refer to ω as *finitely correlated state* generated by (T, ρ) in the sense of [FNW92].

Now, we consider the correlations for the functional $\tilde{\omega}(T, \rho)$ on the infinite spin chain, if we choose the generating channel T as a k -th root of the bistochastic CDC, that is $\mathbb{E}_{\mathbb{1}}^k(X) = \frac{1}{d} \text{tr} X$, and ρ as the maximally mixed state, i.e. $\rho = \frac{1}{d}$. Then we find that the functional splits into two independent parts, if we ignore the measurements on k adjacent

spins:

$$\begin{aligned}
& \tilde{\omega} \left(A_{[-\infty, n)} \otimes \underbrace{\mathbb{1} \otimes \dots \otimes \mathbb{1}}_{k\text{-times}} \otimes A_{[n+k, \infty]} \right) \\
&= \text{tr} \left(\rho E_{A_{-\infty}} \circ \dots \circ E_{A_{n-1}} \circ E_{\mathbb{1}}^k \circ E_{A_{n+k}} \circ \dots \circ E_{A_{\infty}}(\mathbb{1}) \right) \\
&= \text{tr} \left(\rho E_{A_{-\infty}} \circ \dots \circ E_{A_{n-1}}(\mathbb{1}) \right) \text{tr} \left(\rho E_{A_{n+k}} \circ \dots \circ E_{A_{\infty}}(\mathbb{1}) \right) \\
&= \tilde{\omega} \left(A_{[-\infty, n)} \right) \tilde{\omega} \left(A_{[n+k, \infty]} \right). \tag{6.28}
\end{aligned}$$

Since $\mathcal{A}$ is a quasi-local algebra, this $\tilde{\omega}$ is well-defined. The state $\tilde{\omega}$ is called a k -dependent state. Thus the maximal roots generate k -dependent states on an infinite spin chain, where the correlation length is $d^2 - 1$, with d the dimension of ρ . If we consider the construction in (6.22) we find that actually the maximal roots define the maximal finite correlation length for k -dependent states, where the dimension of each element of the algebra $\mathcal{A}$ coincides with the Kraus rank of the root channel T .

6.2.2 Strictly Forgetful Memory Channels

The theory of *quantum memory channels* has been first set up in [KW05]. Before this theory was established, the investigations on the concatenation of quantum channels were restricted by the assumption that successive channel inputs act independently, which was not sufficient to describe many real-world experiments.

The basic idea of a quantum channel with memory is to add an additional system $\mathcal{M}$ to the input and output system of a channel, which we refer to as the *memory system*. Figure 6.2 illustrates the setting. For the sake of clarity we define $\mathcal{B}(\mathcal{H}_{in}) =: \mathcal{A}$ for the

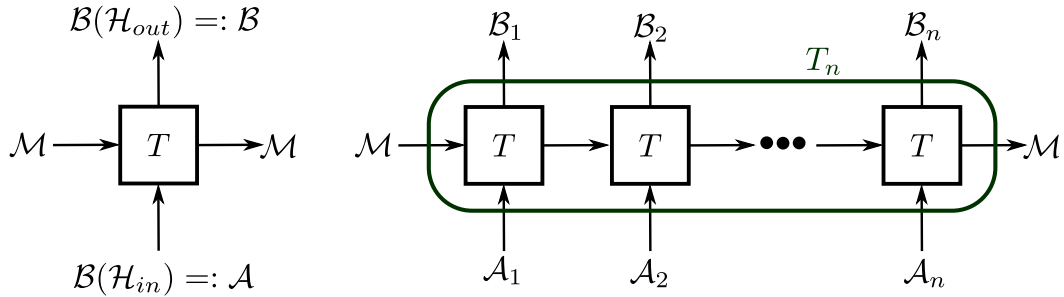


Figure 6.2: A quantum memory channel T and its n -fold concatenation T_n . $\mathcal{M}$ denotes the memory systems which is used to model interaction between the different concatenation steps.

input system and $\mathcal{B}(\mathcal{H}_{out}) =: \mathcal{B}$ for the output system. Then, the n -fold concatenation $T_n : \mathcal{B}^{\otimes n} \otimes \mathcal{M} \mapsto \mathcal{M} \otimes \mathcal{A}^{\otimes n}$ of the quantum memory channel $T : \mathcal{B} \otimes \mathcal{M} \mapsto \mathcal{M} \otimes \mathcal{A}$ can be expressed via:

$$T_n = \left(T \otimes id_{\mathcal{A}}^{\otimes n-1} \right) \circ \dots \circ \left(id_{\mathcal{B}}^{\otimes n-2} \otimes T \otimes id_{\mathcal{A}} \right) \circ \left(id_{\mathcal{B}}^{n-1} \otimes T \right), \tag{6.29}$$

where id denotes the ideal channel. Due to this construction, the elements of the output algebras $\mathcal{B}_i$ will certainly be affected by the choice of the initial memory state ρ of the memory input system $\mathcal{M}$. But the question arises, if the memory initializer influences all elements of the output algebra in the same way. This introduces the concept of *forgetful* memory channels. This is the class of memory channels, where the effect of the initializing memory becomes arbitrarily small for a suitable number of channel concatenations. More formally, we define a forgetful channel in the following way:

Definition 6.1 (Forgetful Quantum Channel [Sch05]). Let $T : \mathcal{B} \otimes \mathcal{M} \mapsto \mathcal{M} \otimes \mathcal{A}$ be a quantum memory channel, T_n its n -fold concatenation and let $\hat{T}_n : \mathcal{M} \mapsto \mathcal{M} \otimes \mathcal{A}^{\otimes n}$ be the concatenated channel where the outputs are ignored: $\hat{T}_n(M) := T_n(\mathbb{1}_{\mathcal{B}}^{\otimes n} \otimes M)$ for all $M \in \mathcal{M}$. Furthermore, let P denote the CDC. Then T is called **forgetful** if and only if for all $\varepsilon > 0$ there is a $n_0 \in \mathbb{N}$, such that for all $n \geq n_0$:

$$\left\| \hat{T}_n - (P \otimes id_{\mathcal{A}}^{\otimes n}) \circ \hat{T}_n \right\|_{cb} \leq \varepsilon \quad (6.30)$$

and thus $\lim_{n \rightarrow \infty} \left\| \hat{T}_n - (P \otimes id_{\mathcal{A}}^{\otimes n}) \circ \hat{T}_n \right\|_{cb} = 0$.

$\|T\|_{cb}$ denotes the norm of *complete boundness*, which we will not further explain in this thesis (for further information see [Pau03]). Different criteria for forgetful quantum

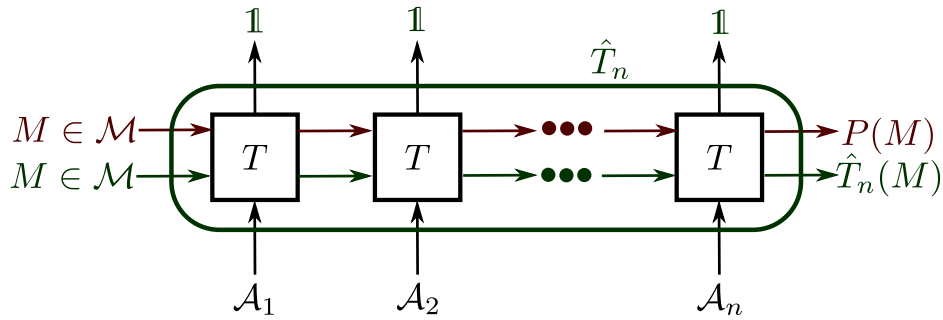


Figure 6.3: The channels $\hat{T}_n$ and $(P \otimes id_{\mathcal{A}}^{\otimes n}) \circ \hat{T}_n$ just differ in the behavior on the memory system. From the definition 6.30 we see that T is forgetful if and only if the cb -norm of the difference between both channels becomes arbitrary small for large n .

channels were recently investigated in [Sal10]. In order to put the definition of forgetfulness into the context of finite roots of the CDC, we adapt the definition to a finite number of concatenation steps. With the same notations as in definition 6.30, we define:

Definition 6.2 (Strictly Forgetful Memory Channel). A quantum memory channel T is **strictly forgetful** if and only if there is an $n \in \mathbb{N}$ such that

$$\left\| \hat{T}_n - (P \otimes id_{\mathcal{A}}^{\otimes n}) \circ \hat{T}_n \right\|_{cb} = 0. \quad (6.31)$$

We refer to the minimal n , such that this equation holds, as *memory depth* of the channel T .

If we reformulate these definitions for a channel S *without* input algebra $\mathcal{A}$, that is $S : \mathcal{B} \otimes \mathcal{M} \mapsto \mathcal{M}$, we find $\hat{S} : \mathcal{M} \mapsto \mathcal{M}$, $\hat{S}(M) := S(\mathbb{1}_{\mathcal{B}} \otimes M)$. Moreover, (6.31) equals

then:

$$\left\| \hat{S}_n - P \right\|_{cb} = 0. \quad (6.32)$$

Finally, the channel $\hat{S}$ can be defined as the Stinespring representation of a maximal root R , i.e. $\hat{S}(M) := V^*(\mathbf{1} \otimes M)V$ with V the defining isometry of R as in (6.19). With this definition of $\hat{S}(M)$ it follows that:

$$\left\| \hat{S}_{d^2-1} - P \right\|_{cb} = \|P - P\|_{cb} = 0. \quad (6.33)$$

In conclusion, we find that a maximal qubit root R defines a strictly forgetful memory channel S , without input system $\mathcal{A}^{\otimes n}$. The memory depth of S is then $d^2 - 1$ with d as the dimension of the memory system. Further investigations are necessary to determine the behavior of the generated strictly forgetful channel, if we connect some input.

CHAPTER 7

Summary and Outlook

In the present thesis we investigated the divisibility of completely depolarizing channels. For this purpose, we introduced the mathematical concept of quantum channels and focussed our attention to channels with the same input and output dimensions represented by quadratic matrices. Utilizing the theorem of Jordan decomposition we found an upper bound on the order of a root. The essential objective of this thesis was to find a way to construct roots that are as close as possible to that bound. To this end we considered different classes of channels.

In a first step we investigated classical systems as a pre-stage of quantum systems and characterized the entire set of roots for an arbitrary three dimensional classical system. Moreover, we introduced a method to construct roots of maximal order *independent* from the dimension of the system and presented a way to embed these roots into quantum systems. Starting from the decomposition of a bistochastic qubit channel into a unitary and Pauli diagonal channel, we established a characterization of maximal qubit roots. In addition to that, we investigated possible embeddings of maximal qubit roots into larger systems but found, that the order of the resulting roots was below the order of the classical embedded roots. To reach beyond the order of classical embeddings we considered Clifford channels with prime input and output dimension. We estimated an upper bound on the order of a Clifford root, which is asymptotically equivalent to the maximal order, and proved that it can be reached. As a main result we presented a method for the construction of maximal roots in arbitrary dimensions by perturbing the depolarizing channel with an ε -weighted Jordan block.

Finally we point out applications of maximal roots to finitely correlated states and quantum memory channels. In the case of the finitely correlated states, it turns out that the maximal order of a root is also the maximal finite correlation length. Moreover, a maximal root defines a strictly forgetful quantum channel with memory depth $d^2 - 1$.

As an outlook we refer to the open question of minimal Kraus operators to construct a maximal root. So far, we have only been able to answer this question only in the qubit case. Moreover, the aspect of the input for the behavior of the strictly forgetful quantum channels, generated by a maximal root, raises interesting issues. The evolution of quantum memory channels in dependency of their input was recently investigated in [Sal10], but only forgetful, not strictly forgetful, channels were considered.

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Certification/Erklärung

Hiermit versichere ich, dass ich diese Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel verwendet habe.

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